



ARTIFICIAL INTELLIGENCE

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ML Supervised Learning

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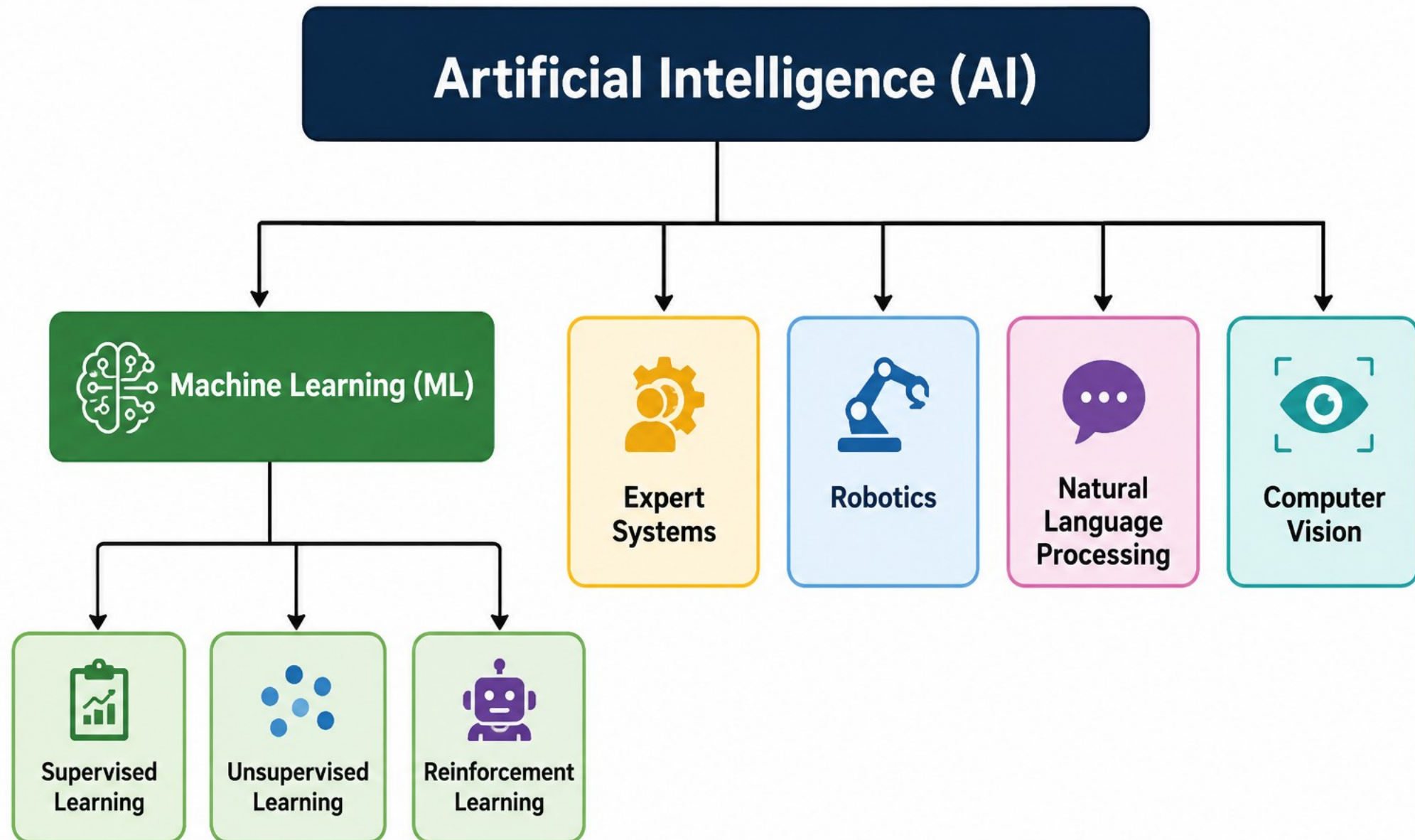
ML

- Artificial Intelligence (AI) is a broad field of computer science that focuses on creating machines capable of performing tasks that normally require human intelligence, such as reasoning, learning, problem-solving, perception, and decision-making.
- However, building an AI system by manually programming rules (hard coded) for every possible situation is difficult and often impossible.
- As the amount of data and complexity of problems increased, researchers needed a way for computers to learn automatically from experience instead of being explicitly programmed.
- **This led to the development of Machine Learning (ML)**

Definition of Machine Learning

Machine Learning is a branch (subset) of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed.

Artificial Intelligence aims to make machines intelligent, while Machine Learning is one of the methods used to achieve AI by enabling computers to learn from data rather than relying solely on predefined rules.



Real-Life Examples

Artificial Intelligence

Self-driving cars

Voice assistants

Email assistant

Netflix recommendations

Medical diagnosis

Machine Learning

Learning road signs from images

Speech recognition models

Spam email detection

Predicting movies you may like

Disease prediction from patient data

Machine Learning is needed because it enables systems to:

Learn from Data – Automatically discover patterns instead of relying on manually written rules.

Adapt Automatically – Improve performance as more data becomes available.

Improve with Experience – Become more accurate over time through training.

Handle Complex Problems – Solve tasks where writing explicit rules is difficult or impossible.

Generalize to New Data – Make correct predictions for previously unseen examples.

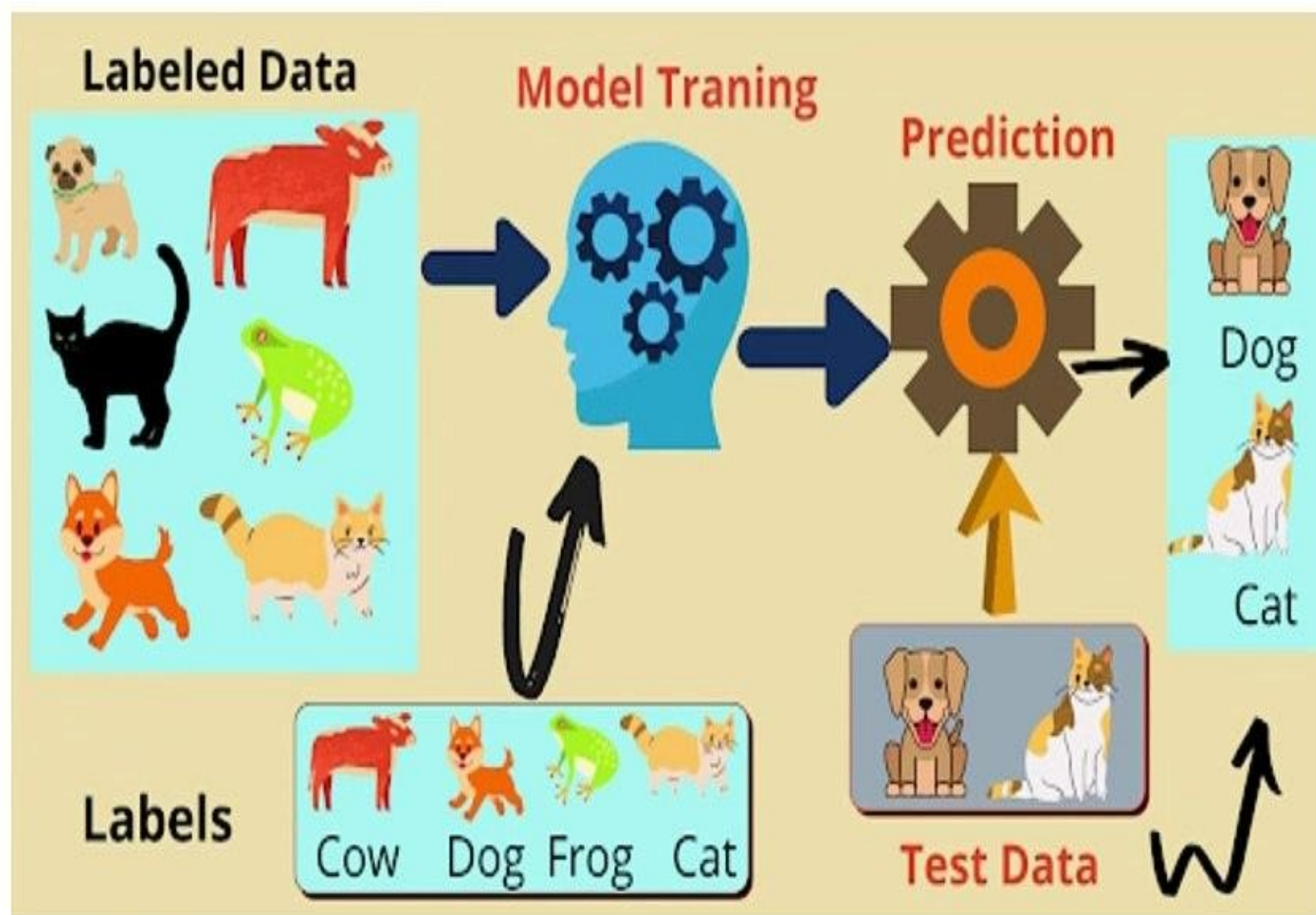
Types of Machine Learning (ML)

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

Supervised Learning

- Supervised Learning is a task-driven learning approach in which the machine learns from labeled data.
- Each training example contains:
- Input (Features)
- Correct Output (Label)
- The goal is to learn the relationship between the input and output so that the model can predict the correct output for new, unseen data.

SUPERVISED LEARNING



the model **learns** from the **label**

Labelled Data

Cat



Dog



TRAINING



INPUT
(Test data)



OUTPUT



1. **Input labeled Image:** Cat Image, Dog Image

2. **The machine studies thousands of labeled animal images and learns their characteristics:**

Cats → Pointed ears, whiskers

Dogs → Different face and body structure

Rabbits → Long ears

Birds → Beak and wings

3. **Prediction:** Now the machine receives a new image that it has never seen before.

4. Prediction: **Cat**

Even though the image is new, the model correctly classifies it because it learned from the labeled training data.



Cat

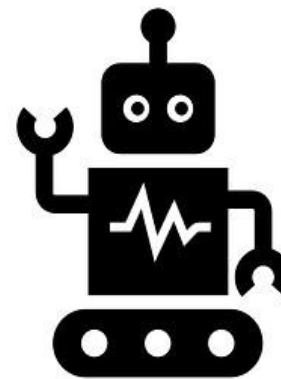


Dog



Bunny

Labelled
pictures



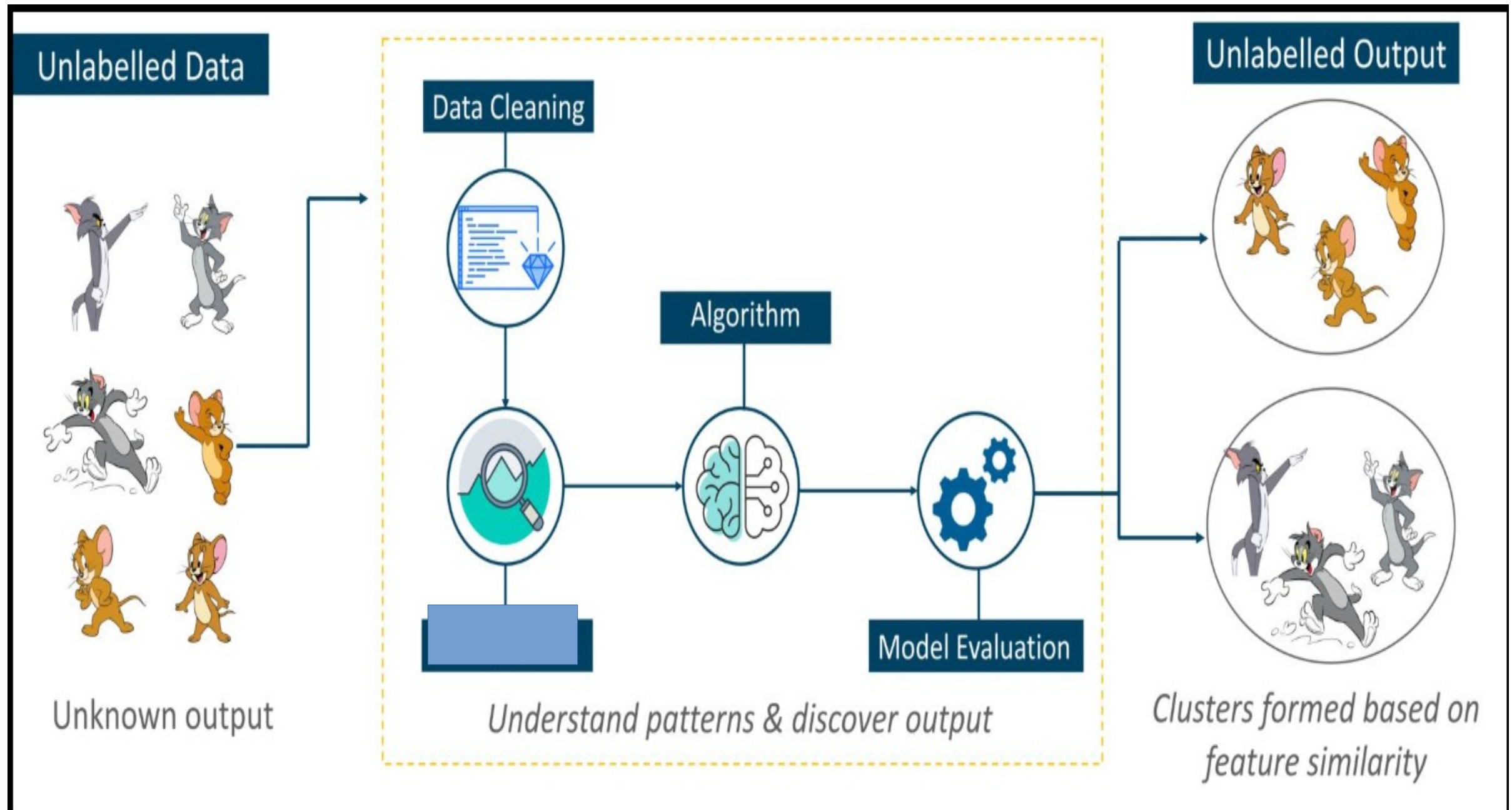
We are **helping machine** to learn by **labeling each picture** with the name of the animal.

Unsupervised Learning

- Unsupervised Learning is a type of Machine Learning in which the algorithm learns from unlabeled data.
- Unlike supervised learning, there are no predefined output labels. The algorithm identifies hidden patterns, structures, or relationships within the data on its own.
- **Working of Unsupervised Learning**
- Collect unlabeled data.
- Preprocess the data.
- Apply an unsupervised learning algorithm.
- The algorithm groups similar data or finds hidden patterns.
- Analyze the results.

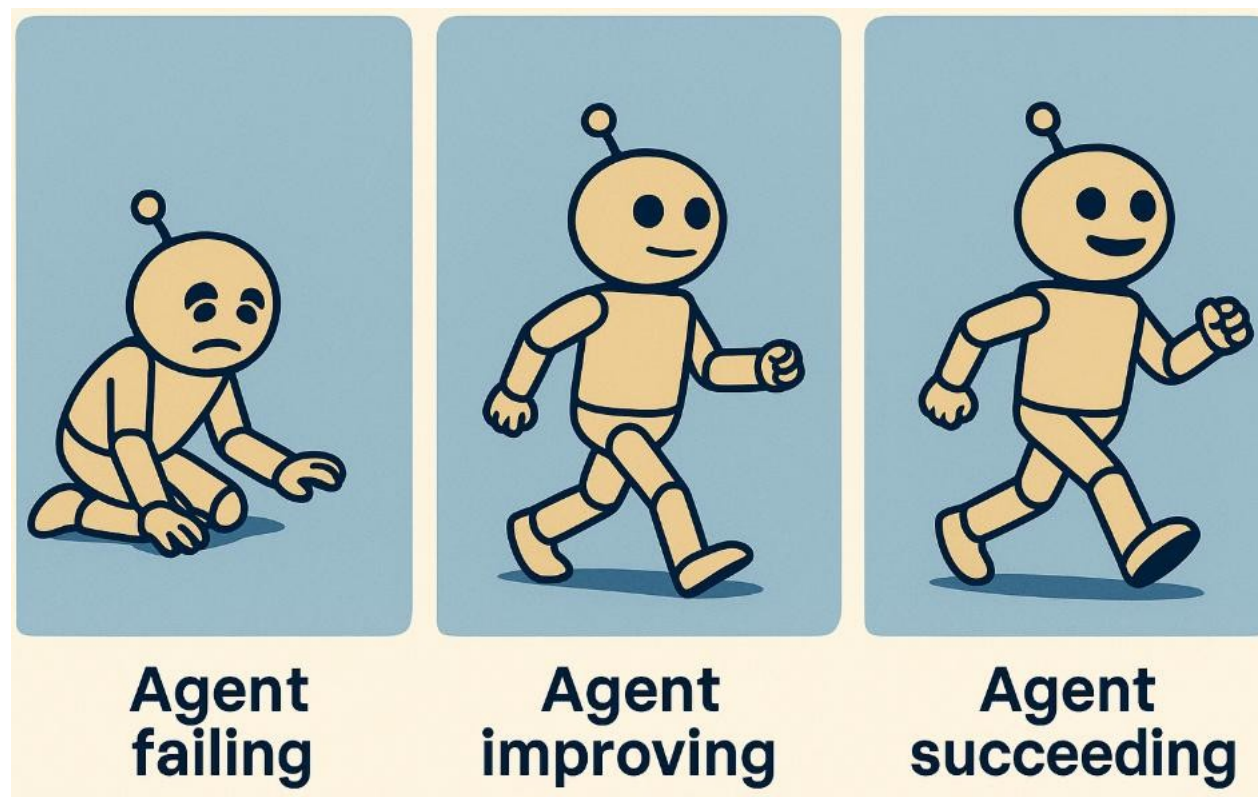
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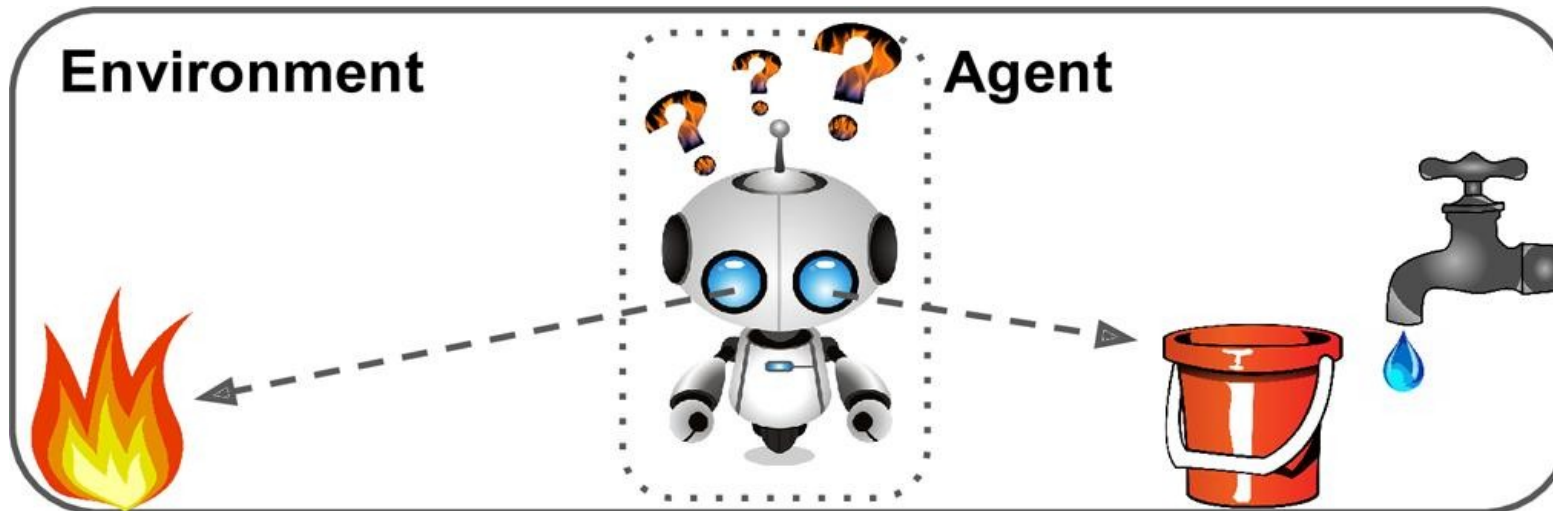
Machine Learning



Reinforcement Learning

- Reinforcement Learning (RL) is a type of Machine Learning in which an agent learns by interacting with its environment. The agent performs actions and receives rewards for correct actions or penalties (punishments) for incorrect actions.
- Over time, the agent learns the best strategy (policy) to maximize the total reward.





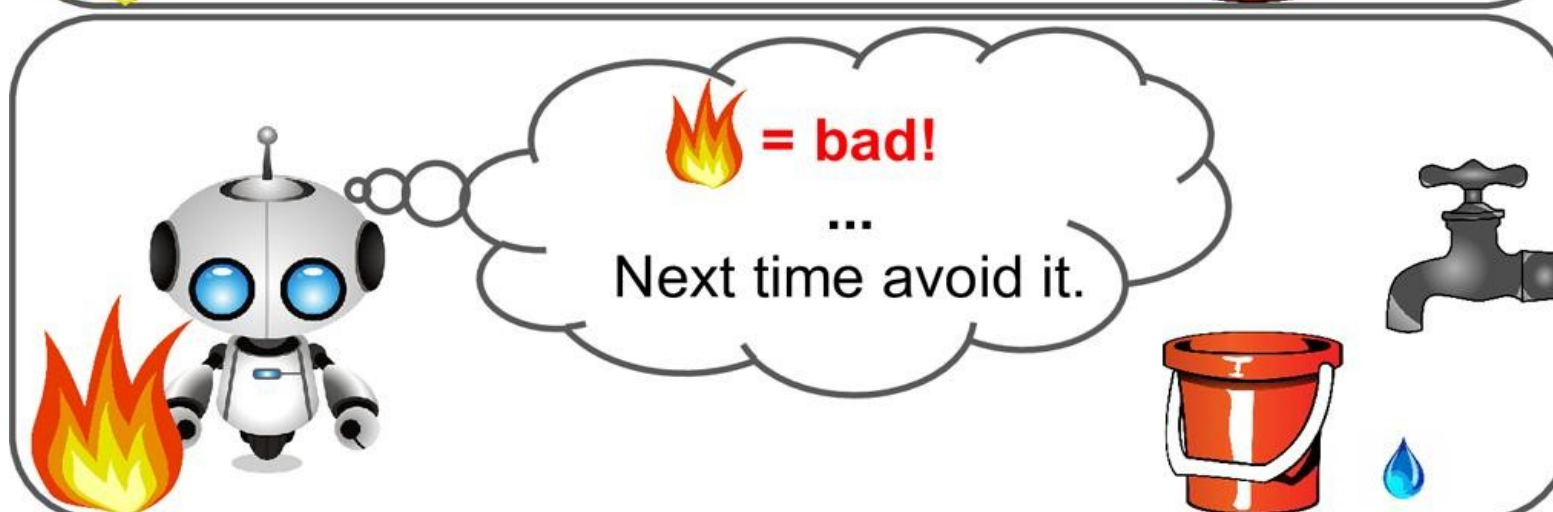
1 Observe

2 Select action using policy



3 Action!

4 Get reward or penalty



5 Update policy (learning step)

6 Iterate until an optimal policy is found

Agent Plays Game



Makes a Move



Wins? —————▶ Yes → Reward (+100)



No → Penalty (-50)



Learns Better Strategy

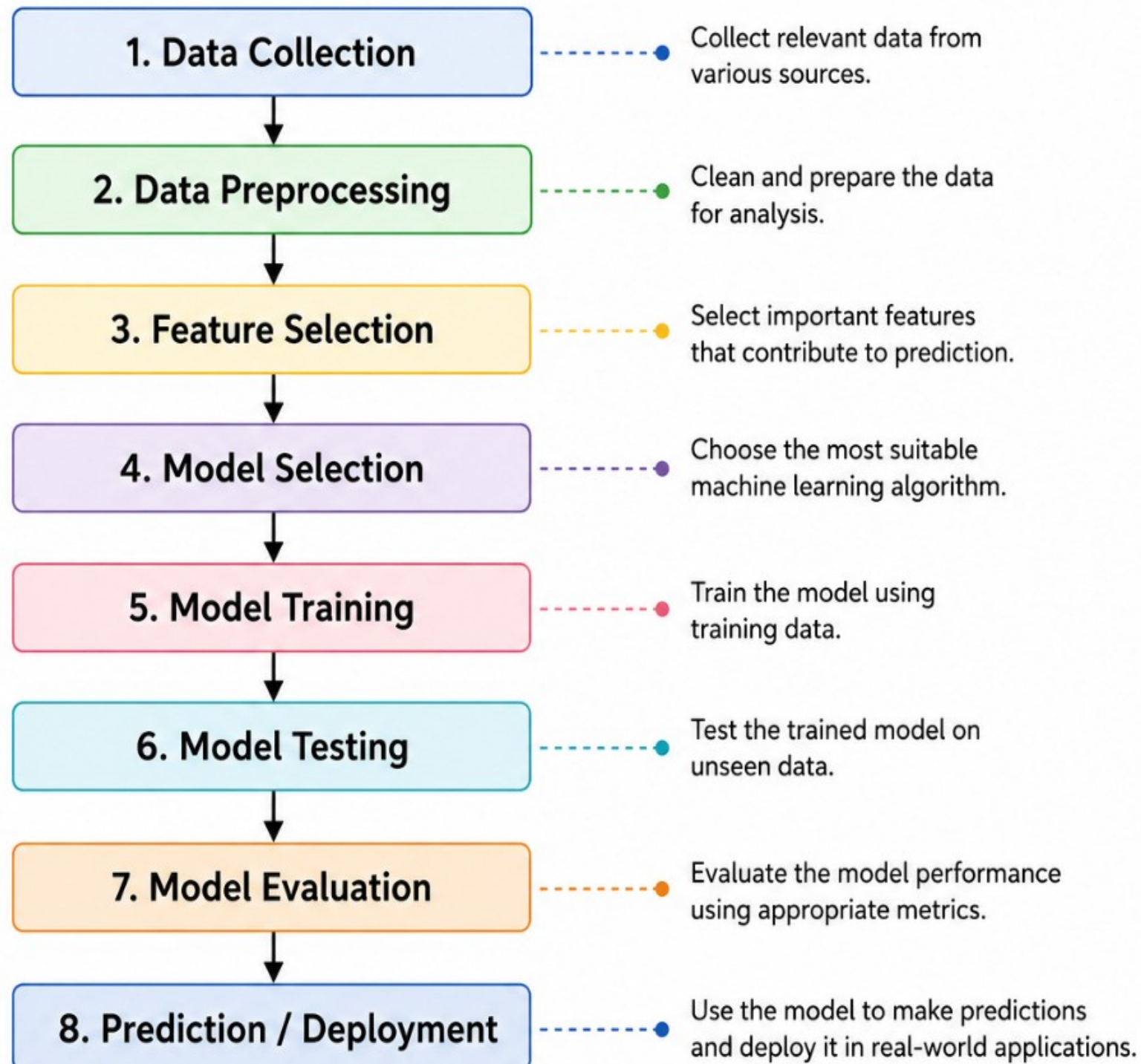
Examples:

Chess AI

Self-driving car simulation

Robot navigation

Machine Learning Process



ML workflow or Objectives

- **Collect Data** – Gather relevant raw data from sources
- **Preprocess Data** – Clean, transform, and prepare data for modeling
- **Split Data** – Divide into training and testing sets
- **Choose Model** – Select an appropriate algorithm for the problem
- **Train Model** – Fit the model to the training data
- **Evaluate Model** – Test performance using metrics like accuracy, RMSE,...
- **Deploy Model** – Use the trained model for real-world predictions

Machine Learning follows a systematic process to build predictive models.

Steps

1. **Data Collection** :Data Collection is the process of gathering relevant data from different sources for training the machine learning model.

Manual Observation ,

Fruit Shops / Supermarkets

Farm

Agricultural Research Centers

Public Datasets (Websites, Databases)

Sensors and Cameras

Collect information about different fruits.

Weight (g)	Color	Diameter (cm)	Fruit
160	Red	8	Apple
180	Yellow	9	Mango
170	Red	8	Apple
220	Yellow	10	Mango

Machine Learning follows a systematic process to build predictive models.

Steps

1. **2. Data Preprocessing:** is the process of cleaning and preparing the collected raw data before training.

2. Remove duplicate records

3. Handle missing values

4. Removing noisy or irrelevant data

5. Correct incorrect values

6. Convert categorical data into numerical values (Red =0, green=1)

3: **Feature Selection :** Feature Selection is the process of choosing the most important input features for prediction.

Weight, Color, Diameter

Machine Learning follows a systematic process to build predictive models.

Steps

4. Model Selection: Model Selection is the process of choosing a suitable machine learning algorithm for the problem.

1. Decision Tree, k-NN, SVM

5. Model Training : During training, the selected algorithm learns patterns from the training dataset.

6: Model Testing: The trained model is tested using unseen data to check how well it performs.

Testing Data :

	Weight	Color	Diameter	Actual Fruit
	165	Red	8	Apple
1	205	Yellow	9	Mango

Machine Learning follows a systematic process to build predictive models.

Steps

7. Prediction:

Actual	Predicted
Apple	Apple
Mango	Mango

1.

8. Model Evaluation:

Model Evaluation measures the performance of the trained model using evaluation metrics.

9. Deployment :

The fruit shop can now automatically classify new fruits.

Machine Learning follows a systematic process to build predictive models.

Disease Prediction:

Collect patient records.

Remove missing values., Handle duplicate records , Normalize data

Select features like age, BP, sugar.

Train Decision Tree.

Test using unseen data.

Deploy in hospitals.

Machine Learning is revolutionizing many sectors.

Industry	Application
Healthcare	Disease diagnosis, medical imaging
Banking	Fraud detection, loan approval
Retail	Product recommendation
Agriculture	Crop disease prediction
Manufacturing	Predictive maintenance
Transportation	Self-driving cars
Education	Personalized learning
Entertainment	Movie recommendations



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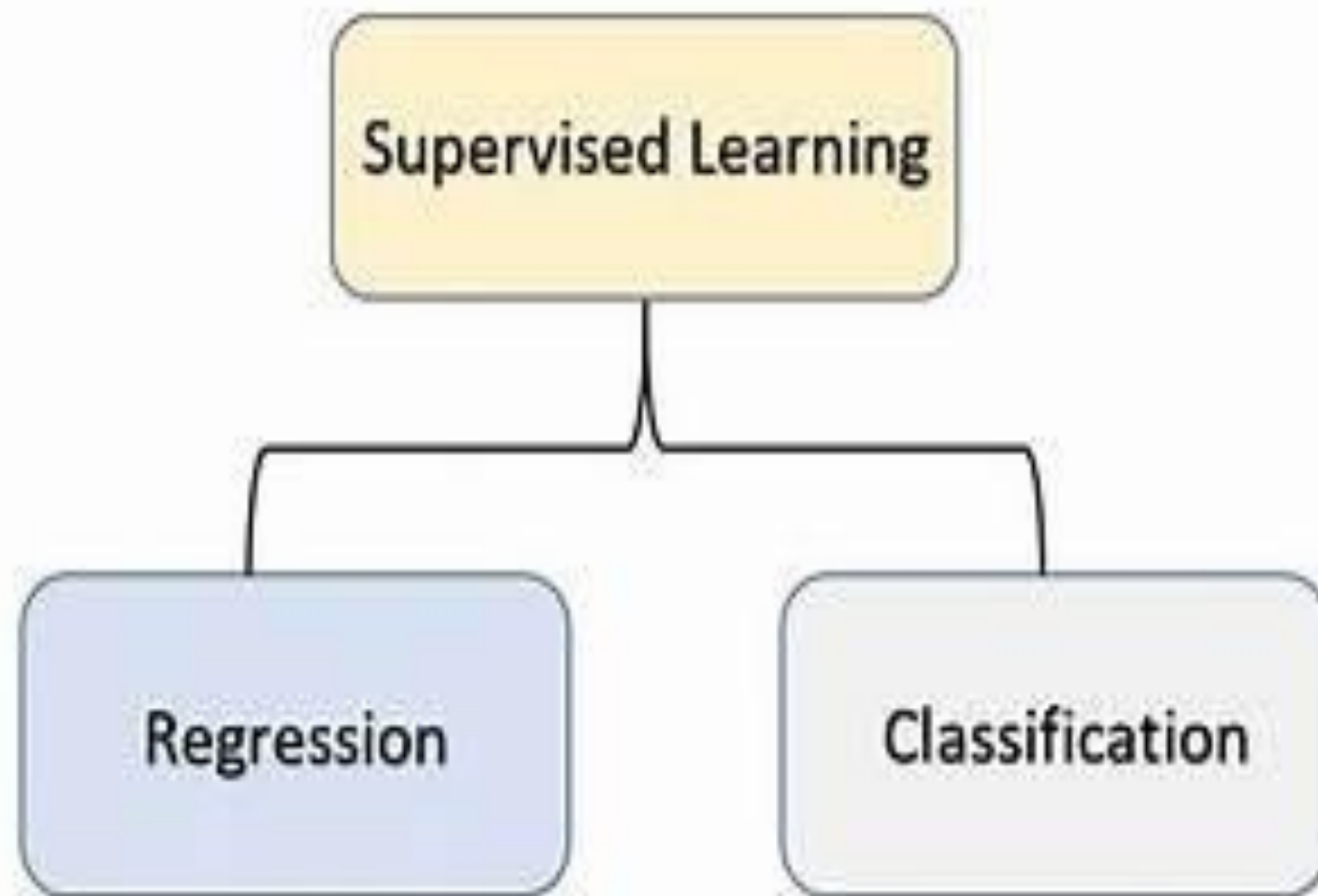
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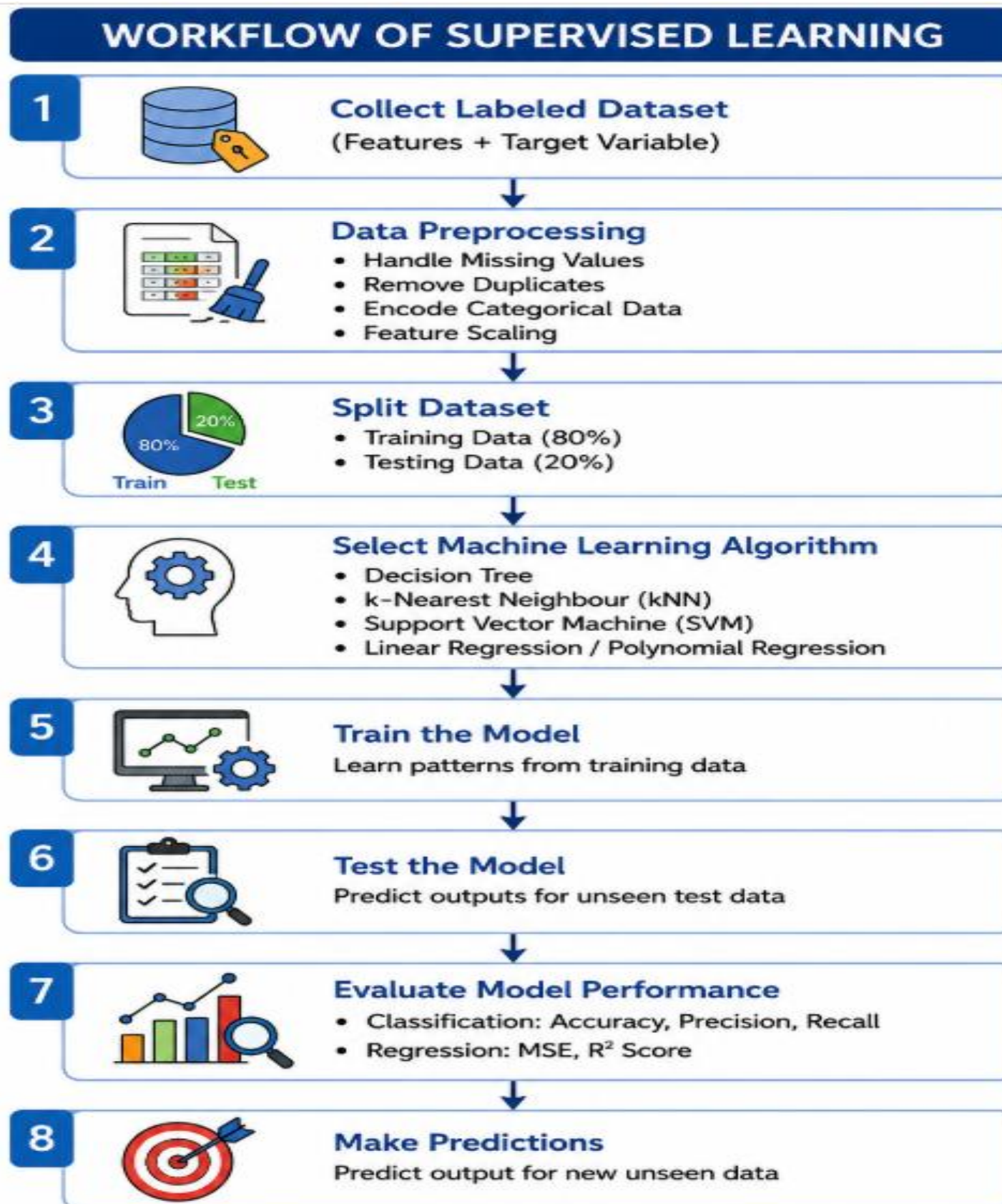
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Supervised Learning is a type of Machine Learning in which the model is trained using labeled data. Each training example consists of input features and a corresponding target (output) value. The objective is to learn the relationship between the inputs and outputs so that the model can accurately predict the output for new, unseen data.

Supervised Learning is broadly divided into two main types based on the nature of the output (target variable).





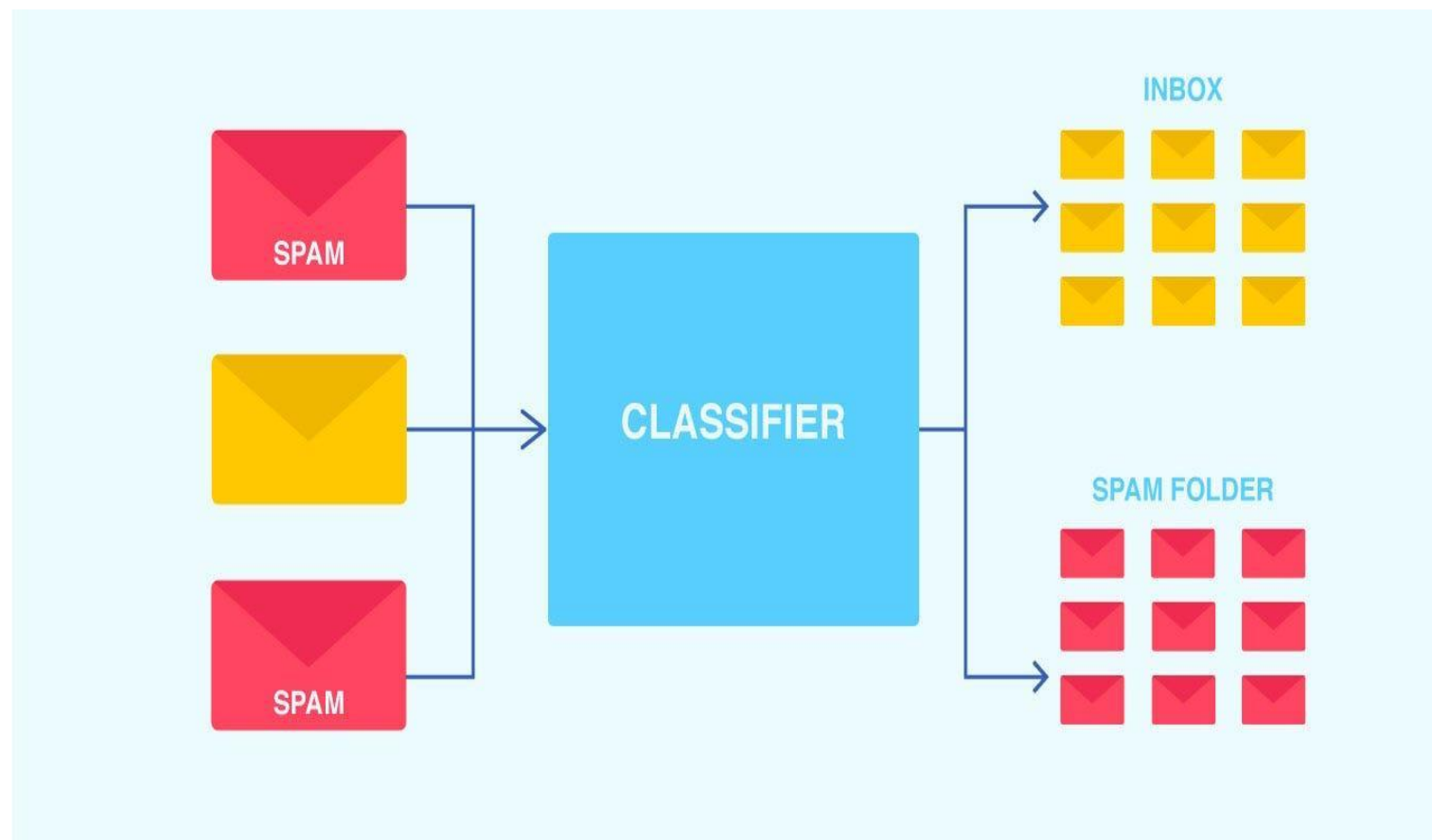
1. Classification

Definition

Classification is a supervised learning task where the output is a category or class label. The model learns from labeled data and predicts which class a new input belongs to.

Examples

- 📧 Spam Email Detection → Spam / Not Spam
- 🐱 Animal Classification → Cat, Dog, Rabbit, Bird
- 🏥 Disease Diagnosis → Diabetic / Non-Diabetic



Classification Uses Discrete (Categorical) Output

A Discrete Variable can take specific, countable values. It cannot have decimal values between two integers.

Examples

Number of Students = 40

Number of Cars = 25

Number of Employees = 150

Number of Books = 12

2. Regression

Regression is a supervised learning task where the output is a continuous numerical value. The model predicts a number rather than a category.

Examples

House Price Prediction → ₹75,00,000

Temperature Prediction → 32°C

Stock Price Prediction → ₹1,250

Salary Prediction → ₹8,50,000/year

Rainfall Prediction → 120 mm

Regression uses A **Continuous Variable** can take any value within a range, including decimal values.

Examples

House Price

Height

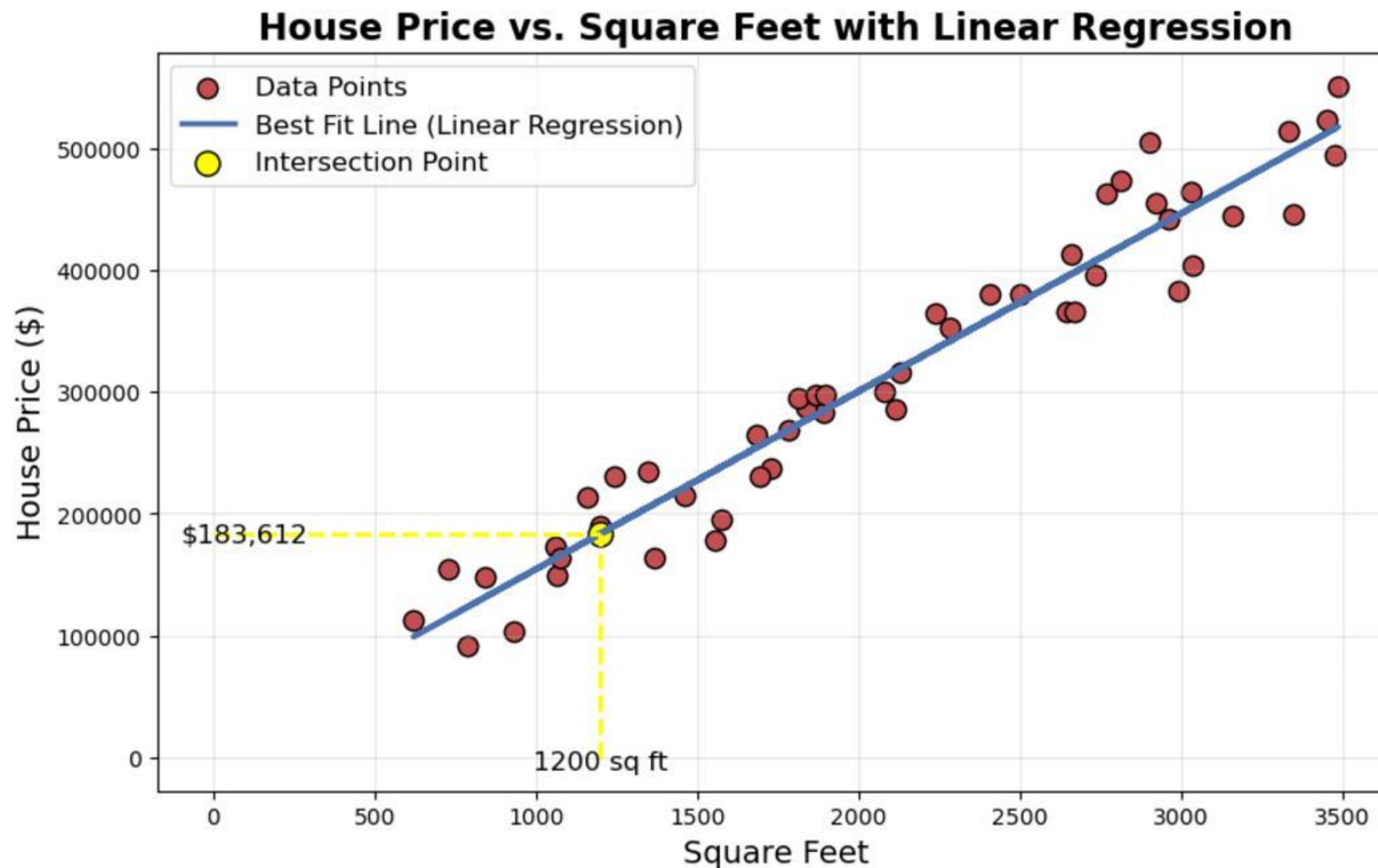
Weight

Temperature

Salary

Distance

Rainfall



the goal is to predict a continuous value (house price) based on the size of the house (square feet).

The house size (independent variable) is used to predict the price (Dependent variable).

Regression analysis is a statistical method for modeling and analyzing the relationship between a dependent variable and one or more independent variables.

Difference between Classification and Regression

Feature	Classification	Regression
Definition	Predicts a category or class label.	Predicts a continuous numerical value.
Output Type	Discrete (Categorical)	Continuous (Numerical)
Target Variable	Discrete Variable	Continuous Variable
Goal	Assign an input to a predefined class or category.	Estimate or predict a numerical value.
Question Answered	"Which class?"	"How much?" or "What value?"
Input-Output Mapping	Maps input (X) to a discrete output (Y).	Maps input (X) to a continuous output (Y).
Output Examples	Spam / Not Spam, Cat / Dog, Pass / Fail	₹25,00,000, 35.5°C, ₹1,250

Difference between Classification and Regression

Real-World Examples	Email Spam Detection, Animal Classification, Disease Diagnosis	House Price Prediction, Temperature Prediction, Salary Prediction
Common Algorithms	Decision Tree, k-NN, SVM, Logistic Regression	Linear Regression, Decision Tree Regression, Random Forest Regression
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score	MAE, MSE, RMSE, R ² Score
		Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R ² Score

"Where does the model get this labeled data?"

Dataset

- A dataset is simply a collection of data used to train and test a Machine Learning model.
- Think of it as a spreadsheet.
- Rows = Patients
- Columns = Information about each patient

FEATURES (X)

Weight(g)	Color	Diameter(cm)	Fruit Name (LABEL)
160	Red	8	Apple
180	Yellow	9	Mango
170	Red	8	Apple
220	Yellow	10	Mango
165	Red	8	?

↑
Testing Data

First 4 rows → Training Data

Model learns: If Color = Red and Diameter ≈ 8 cm

↓
Fruit = Apple

Prediction

Weight =165
Color = Red
Diameter =8

↓

Fruit = Apple

Features (Input Variables)

- The attributes/independent variables used to make predictions
- The variables can be:
 - Numerical: age, oxygen level
 - Categorical: contact risk (low/medium/high), fever (yes/no)

Label / Target (Output Variable)

- The model tries to learn how these features lead to the correct label.

Training Data

- Portion of the dataset used to teach the model
- The model learns patterns (rules) from this data

Testing Data

- Portion of the dataset kept aside to check the model's performance
- Ensures the model can handle new/unseen data

Model

- The mathematical/algorithmic structure that learns patterns from data
- Example: Decision Tree, k-NN, SVM, Logistic Regression

Prediction

- The model's guess about the target for a new data point

Performance metrics

- How often the model's predictions are correct

Overfitting

- When the model memorizes training data instead of learning patterns
- Works perfectly on training data but fails on new data

Underfitting

Underfitting occurs when a machine learning model is too simple to learn the underlying patterns in the training data. As a result, it performs poorly on both the training data and new (testing) data.



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Age	Fever	Cough	Sore Throat	Contact Risk	Oxygen	COVID Positive
25	1 (Yes)	1 (Yes)	0 (No)	High	92	1 (Positive)
40	0 (No)	1 (Yes)	1 (Yes)	Low	97	0 (Negative)
35	1 (Yes)	0 (No)	1 (Yes)	Medium	96	0 (Negative)
60	1 (Yes)	1 (Yes)	0 (No)	High	89	1 (Positive)
50	0 (No)	0 (No)	0 (No)	Low	98	0 (Negative)

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EL- DT : Disease Prediction



Features						Label
Age	Fever	Cough	Sore Throat	Contact Risk	Oxygen	COVID Positive
25	1 (Yes)	1 (Yes)	0 (No)	High	92	1 (Positive)
40	0 (No)	1 (Yes)	1 (Yes)	Low	97	0 (Negative)
35	1 (Yes)	0 (No)	1 (Yes)	Medium	96	0 (Negative)
60	1 (Yes)	1 (Yes)	0 (No)	High	89	1 (Positive)
50	0 (No)	0 (No)	0 (No)	Low	98	0 (Negative)

Train data

Test data

Model : Learn from data

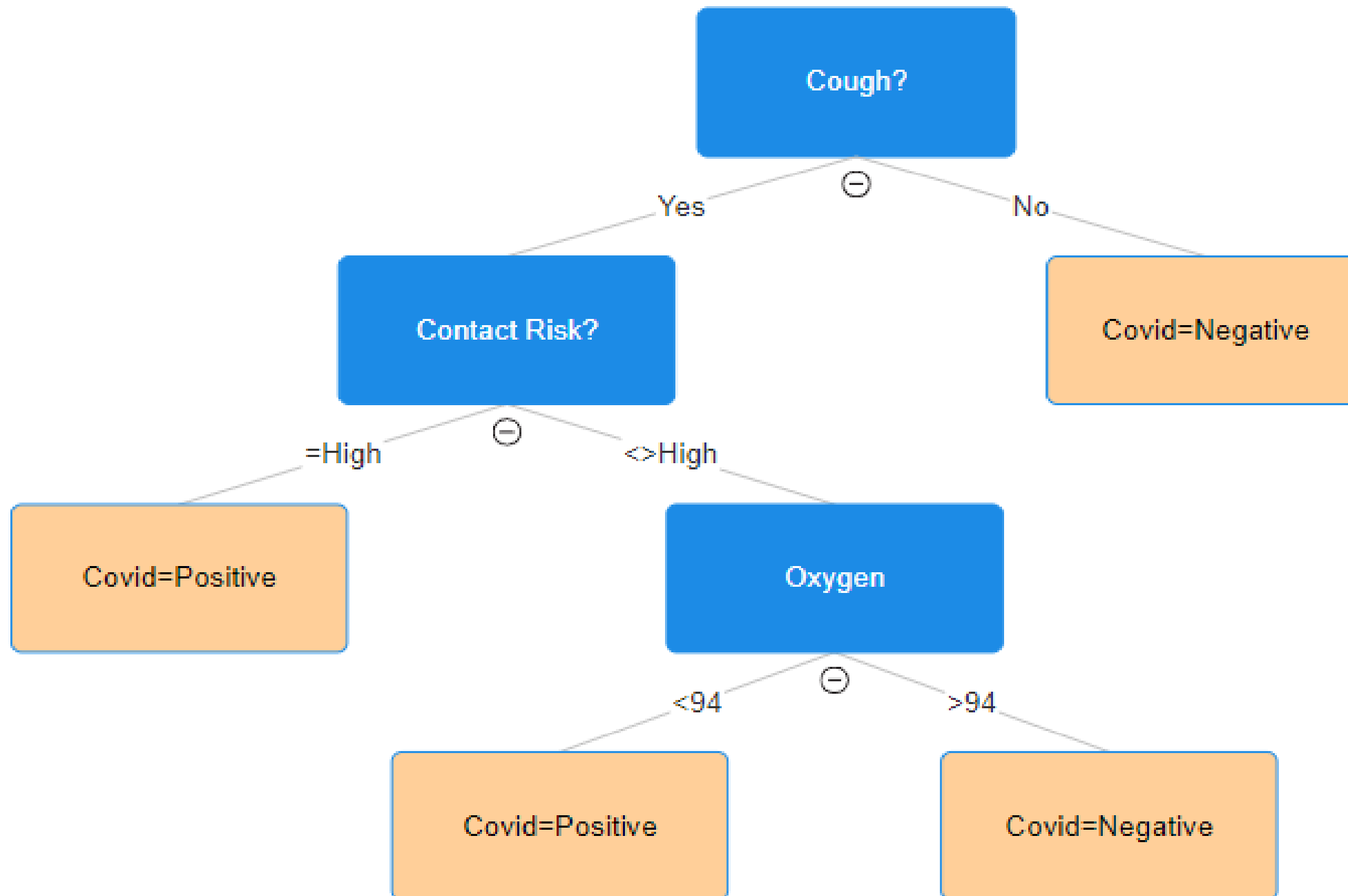
If cough =Yes & Contact Risk=high, then covid=Positive

Prediction

Age=30, Fever=yes, cough=yes, sore throat=no, contact risk=medium, oxygen=88 , then covid=?

Numerical Feature

Categorical Feature



Test sample:

Patient: Age 45, Fever = Yes, Cough = Yes,
Oxygen = 92%, Contact Risk = Medium

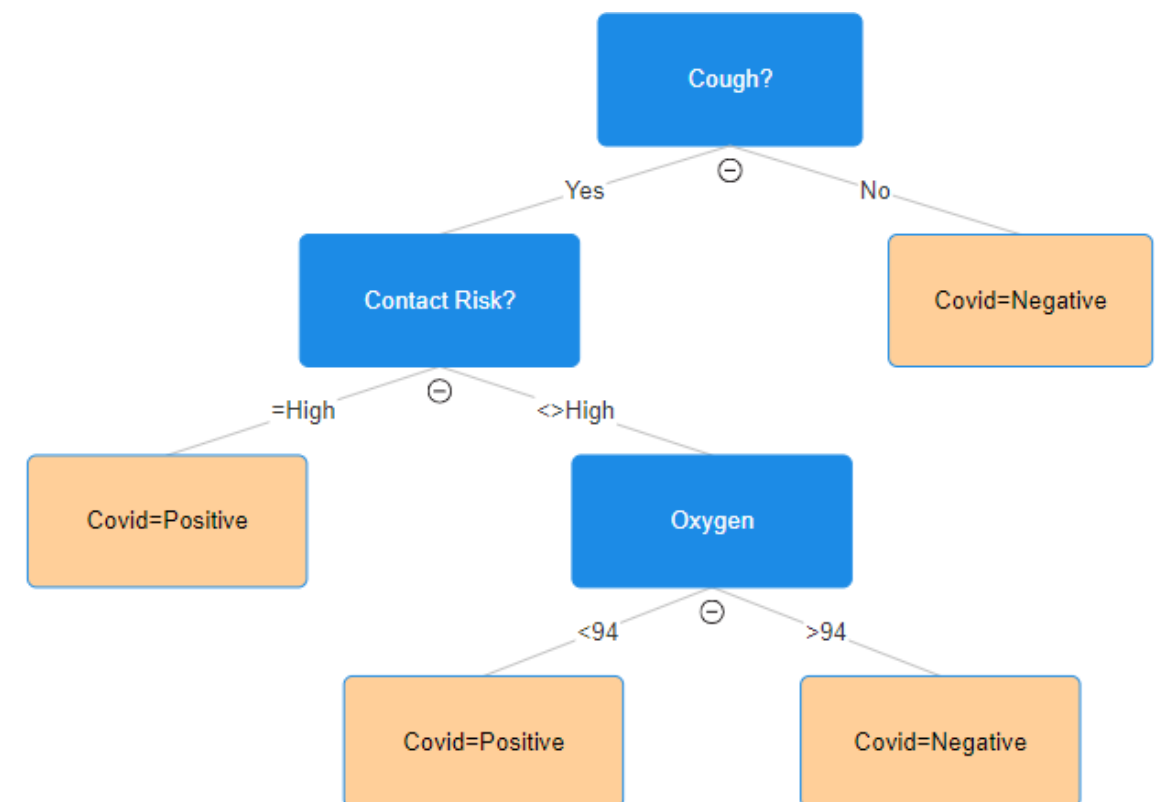
Tree check:

If Cough=yes

->Contact Risk=high ->No

-> Oxygen <94 -> Yes ->

Covid=Positive



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EL- DT : Disease Prediction

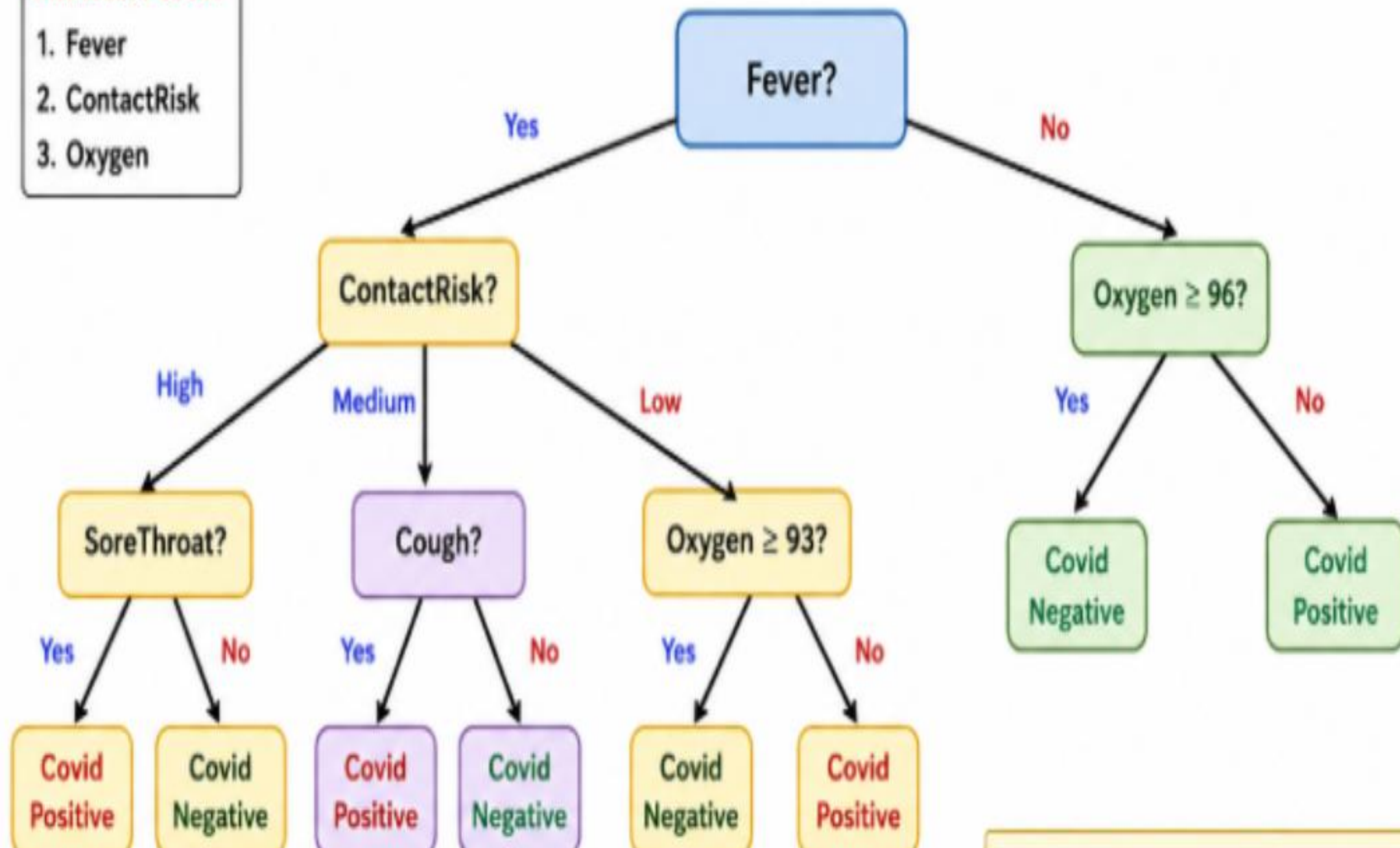
ID	Age	Fever	Cough	SoreThroat	ContactRis k	Oxygen	Covid
T1	65	Yes	Yes	Yes	High	89	Positive
T2	45	Yes	Yes	No	Medium	92	Positive
T3	25	No	No	No	Low	98	Negative
T4	52	Yes	No	No	Low	95	Negative
T5	36	Yes	Yes	Yes	High	90	Positive
T6	29	No	Yes	No	Medium	97	Negative
S1	30	Yes	Yes	No	Medium	88	Positive
S2	70	Yes	No	No	Low	91	Negative
S3	40	No	No	No	High	96	Negative
S4	55	Yes	Yes	Yes	High	94	Positive

DECISION TREE (DT)

(Target Variable: Covid)

Attributes Used:

1. Fever
2. ContactRisk
3. Oxygen



ARTIFICIAL INTELLIGENCE

EL- DT : Disease Prediction

ID	Age	Fever	Cough	SoreThroat	ContactRisk	Oxygen	Covid
T1	65	Yes	Yes	Yes	High	89	Positive
T2	45	Yes	Yes	No	Medium	92	Positive
T3	25	No	No	No	Low	98	Negative
T4	52	Yes	No	No	Low	95	Negative
T5	36	Yes	Yes	Yes	High	90	Positive
T6	29	No	Yes	No	Medium	97	Negative

Take Next feature

Predict=Negative

Both records belong to same class label So it is Pure. Put Label as leaf

Among 4 records of Yes, 3 belong to Positive label and 1 belongs to Negative – So impure. Now Grow this side tree with the next feature. This branch will take only 4 records in the next branch

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EL- DT : Disease Prediction

ID	Age	Fever	Cough	SoreThroat	ContactRisk	Oxygen	Covid
T1	65	Yes	Yes	Yes	High	89	Positive
T2	45	Yes	Yes	No	Medium	92	Positive
T3	25	No	No	No	Low	98	Negative
T4	52	Yes	No	No	Low	95	Negative
T5	36	Yes	Yes	Yes	High	90	Positive
T6	29	No	Yes	No	Medium	97	Negative

=yes

=No

ID	Age	Fever	SoreThroat	ContactRisk	Oxygen	Covid
T1	65	Yes	Yes	High	89	Positive
T2	45	Yes	No	Medium	92	Positive
T5	36	Yes	Yes	High	90	Positive
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ID	Age	Fever	SoreThroat	ContactRisk	Oxygen	Covid
T3	25	No	No	Low	98	Negative
T4	52	Yes	No	Low	95	Negative

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EL- DT : Disease Prediction

=yes

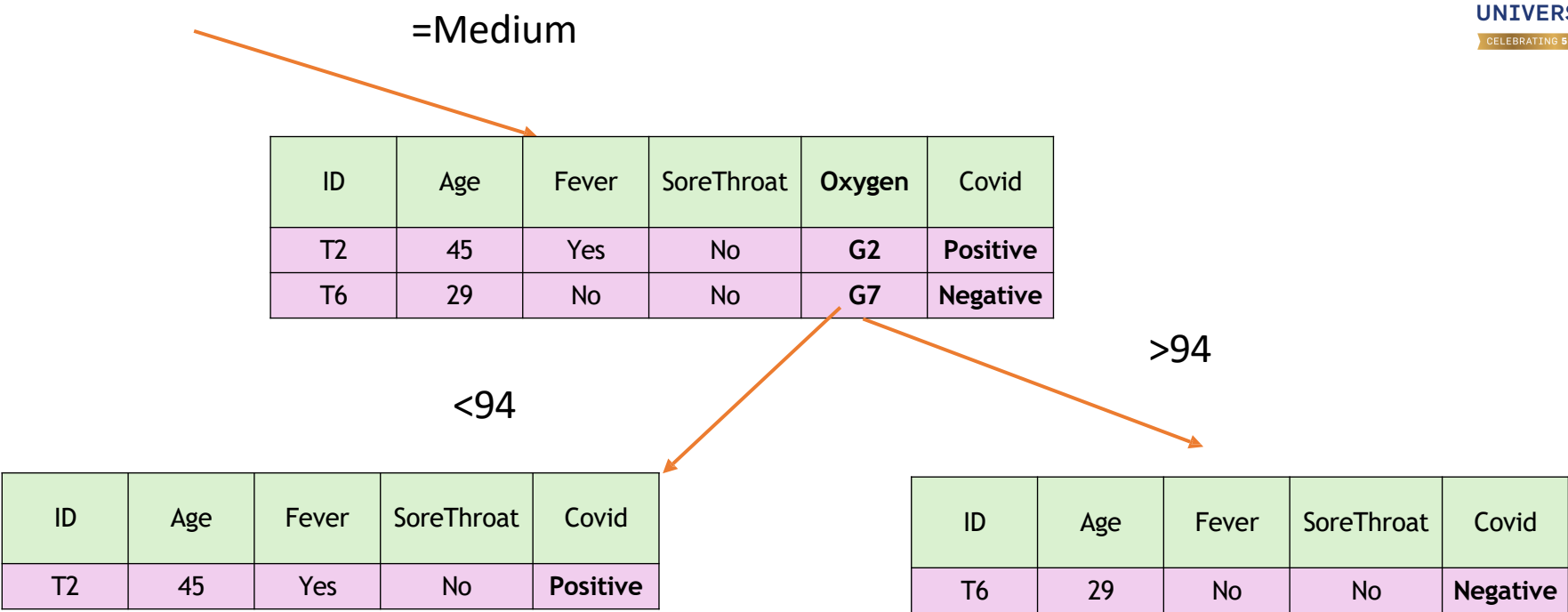
ID	Age	Fever	SoreThroat	ContactRisk	Oxygen	Covid
T1	65	Yes	Yes	High	89	Positive
T2	45	Yes	No	Medium	92	Positive
T5	36	Yes	Yes	High	90	Positive
T6	29	No	No	Medium	97	Negative

=High

ID	Age	Fever	SoreThroat	Oxygen	Covid
T1	65	Yes	Yes	89	Positive
T5	36	Yes	Yes	90	Positive

=Medium

ID	Age	Fever	SoreThroat	Oxygen	Covid
T2	45	Yes	No	92	Positive
T6	29	No	No	97	Negative



ARTIFICIAL INTELLIGENCE

EL- DT : Disease Prediction

ID	Age	Fever	Cough	SoreThroat	ContactRisk	Oxygen	Covid
T1	65	Yes	Yes	Yes	High	89	Positive
T2	45	Yes	Yes	No	Medium	92	Positive
T3	25	No	No	No	Low	98	Negative
T4	52	Yes	No	No	Low	95	Negative
T5	36	Yes	Yes	Yes	High	90	Positive
T6	29	No	Yes	No	Medium	97	Negative
S1	30	Yes	Yes	No	Medium	88	Positive
S2	70	Yes	No	No	Low	91	Negative
S3	40	No	No	No	High	96	Negative
S4	55	Yes	Yes	Yes	High	94	Positive

ID	Age	Fever	Cough	SoreThroat	ContactRisk	Oxygen	Covid
S1	30	Yes	Yes	No	Medium	88	Positive
S2	70	Yes	No	No	Low	91	Negative
S3	40	No	No	No	High	96	Negative
S4	55	Yes	Yes	Yes	High	94	Positive

- For all these 4 records, which we kept aside, has to be used for testing. It is as good as feeding the medicine to the rats at the research lab. If the rat survives, the medicine is good. If not, go back and modify the medicine composition.

Sklearn: means **Scikit-learn**.

It is a popular Python library used for **Machine Learning**.

It provides algorithms and functions for:

- Decision Tree
- KNN
- SVM
- Linear Regression
- Train-test splitting
- Accuracy
- Confusion matrix
- etc.

model_selection is a **module inside Scikit-learn**.

It contains tools used for selecting, evaluating, and preparing machine-learning models.

seaborn

`seaborn` is a **Python visualization library**.

It is mainly used to create attractive statistical graphs, such as:

- Heatmaps
- Bar charts
- Histograms
- Box plots
- Scatter plots

In your program, Seaborn is being used to create the **confusion matrix heatmap**.

```
X_train, X_test, y_train, y_test = train_test_split( X, y,  
test_size=0.30, random_state=42 )
```

The dataset is divided into:

Training data

Used to **teach the Decision Tree.**

Testing data

Used to **check how well the trained tree performs on unseen data.**

```
test_size=0.30
```

This means:

- 70% → training data
- 30% → testing data

If dataset has **16 records**:

$$16 \times 0.30 = 4.8$$

Since we cannot have 4.8 rows, it becomes **5 test samples**.

```
random_state=42 mean?
```

`train_test_split()` randomly selects the records for training and testing.

If we don't specify `random_state`, the split can change every time we run the program.

Variable

`X_train`

`X_test`

`y_train`

`y_test`

Meaning

Features used for training

Features used for testing

Actual classes for training

Actual classes for testing

```
encoder = LabelEncoder()
```

ContactRisk contains text. Machine-learning algorithms need numerical input, so LabelEncoder converts them to numbers.

LabelEncoder determines the mapping from the labels. You can see the actual mapping using:

```
print(encoder.classes_)
```

Select the features

```
X = data.drop("COVID", axis=1)
```

Here we create **X**, which contains the input features.

We remove the COVID column because COVID is the label we want the model to predict.

Select the target

```
y = data["COVID"]
```

y contains the **output/target/class**.

X → Symptoms/features

y → COVID class

Create the Decision Tree

```
model = DecisionTreeClassifier(random_state=42)
```

This creates the Decision Tree model.

Train the Decision Tree

```
model.fit(X_train, y_train)
```

This is where the actual learning happens.

The model receives:

X_train → Features

y_train → Correct answers

The Decision Tree tries to find rules that separate the classes.

Make predictions

```
y_pred = model.predict(X_test)
```

Now we give the **unseen test data** to the trained Decision Tree.



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A Decision Tree is a supervised machine learning algorithm that makes decisions by asking a sequence of questions (yes/No) about the input features.

makes predictions by splitting the data into smaller groups based on a sequence of conditions.

It resembles a tree structure, where each decision is based on a feature, and the final outcome is reached after evaluating these conditions.

Characteristics of Decision Tree

It is a Supervised Learning algorithm.

Can be used for both Classification and Regression.

Represents decisions in a tree-like hierarchical structure.

Easy to understand and interpret.

Works with both **numerical and categorical** data.

Requires minimal data **preprocessing**.

Simple Analogy

1. Doctor's Diagnosis

A doctor does not prescribe medicine after asking only one question. The doctor asks about symptoms, medical history, and test results before reaching a diagnosis.

Likewise, a Decision Tree examines different features before making a decision.

2. Shopping for a Laptop ?

A Decision Tree consists of three main components:

1. Root Node (Tree node)

The topmost node of the tree.

Represents the best feature selected for the first split.

There is only one root node in a decision tree.

2. Decision Node (Internal Node)

Represents a test or condition on a feature.

Splits the data into smaller subsets based on the feature values.

A tree can have multiple decision nodes.

Example: Diameter $>$ 8 cm?

A Decision Tree consists of three main components:

3. Leaf Node (Terminal Node)

Represents the final prediction or decision.

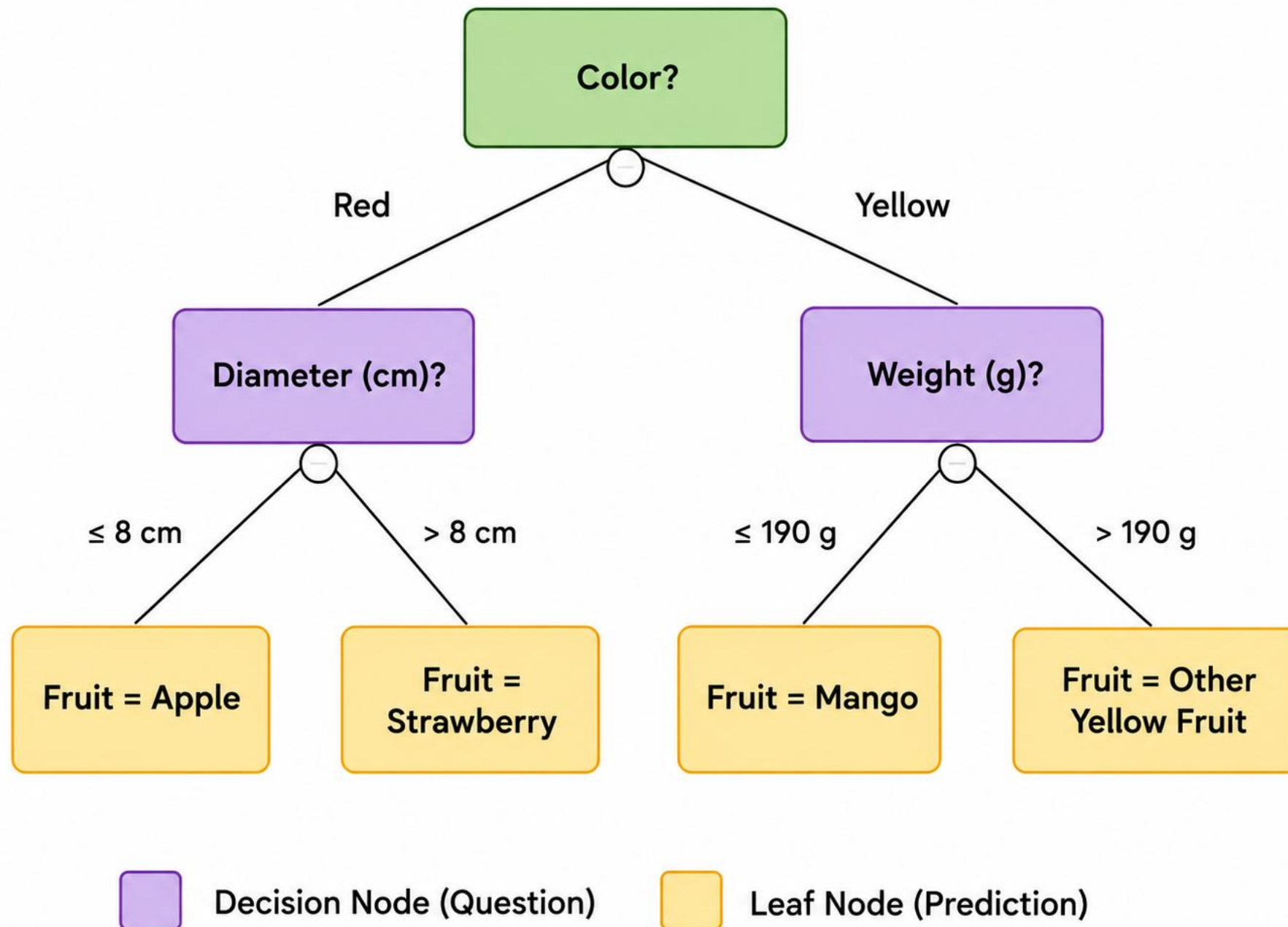
Does not split further.

Contains the class label or predicted value.

Example: Apple or Mango

Training Dataset				
ID	Weight (g)	Color	Diameter (cm)	Fruit
1	160	Red	8	Apple
2	165	Red	8	Apple
3	170	Red	9	Strawberry
4	175	Red	10	Strawberry
5	180	Yellow	9	Mango
6	185	Yellow	10	Mango
7	200	Yellow	11	Other Yellow Fruit
8	220	Yellow	12	Other Yellow Fruit

Example 1



The Decision Tree algorithm learns from the training data and identifies patterns between the **features (Weight, Color, Diameter)** and the target label (Fruit Name).

Based on these patterns, it automatically creates a set of decision rules, which together form the Decision Tree Model.

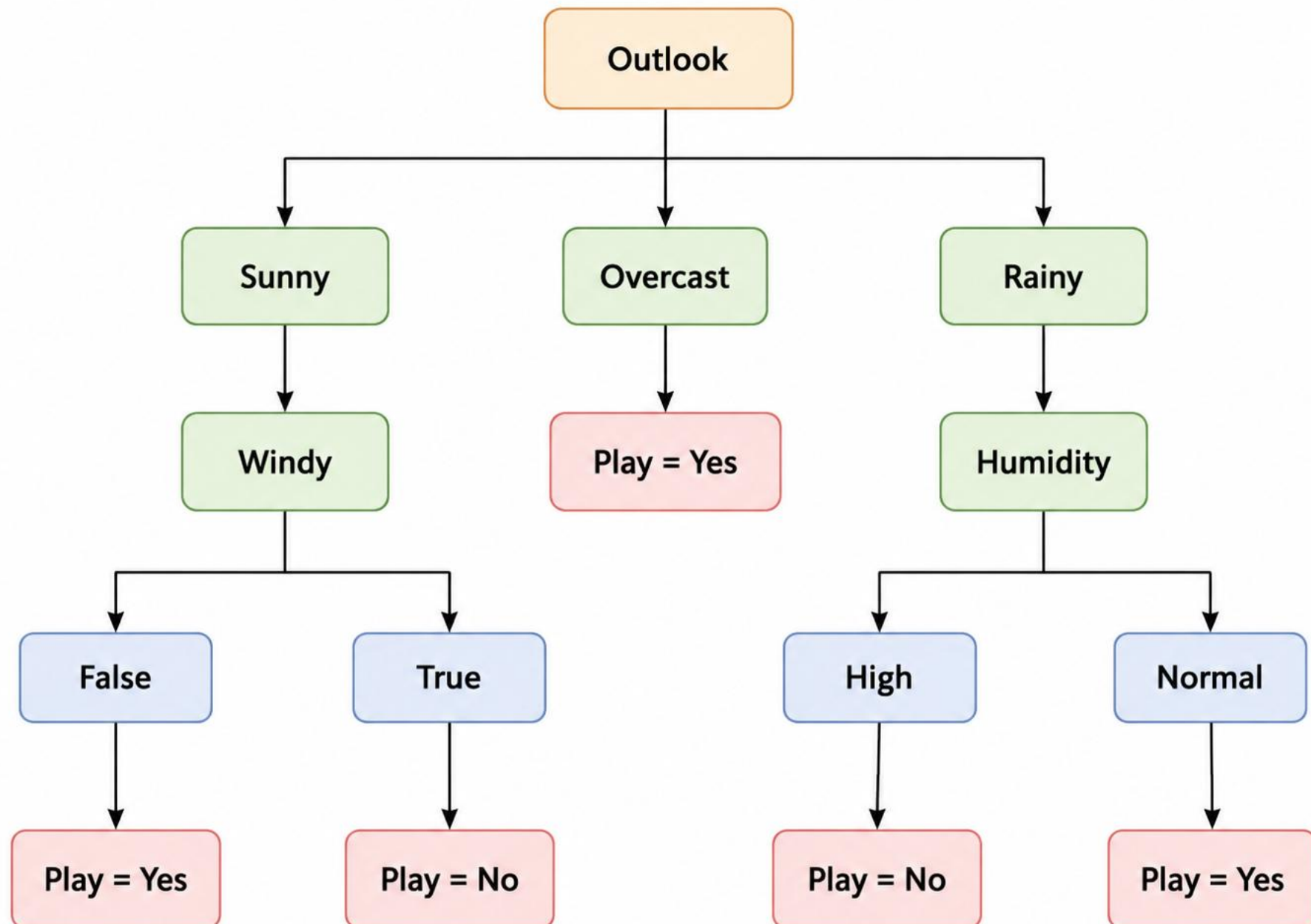
Model: A trained machine learning model that can classify new fruits

Decision Tree Construction Algorithm

1. Start with the complete fruit dataset.
2. Select the most informative feature as the Root Node.
3. Create a decision question using the selected feature.
4. Split the dataset into smaller subsets.
5. Check whether each subset contains only one fruit class.
6. If Yes, create a Leaf Node.
7. Otherwise, repeat the process using the remaining features.
8. Continue until all leaf nodes are pure or no further splitting is possible.

Example 2

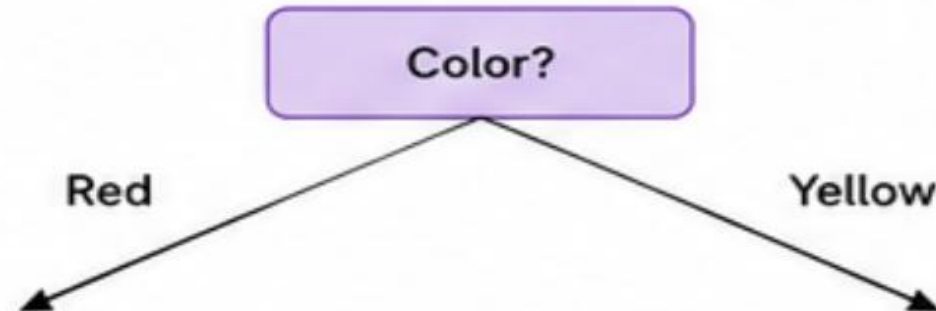
Decision Tree for Play Tennis



Training Dataset				
ID	Weight (g)	Color	Diameter (cm)	Fruit
1	160	Red	8	Apple
2	165	Red	8	Apple
3	170	Red	9	Strawberry
4	175	Red	10	Strawberry
5	180	Yellow	9	Mango
6	185	Yellow	10	Mango
7	200	Yellow	11	Other Yellow Fruit
8	220	Yellow	12	Other Yellow Fruit

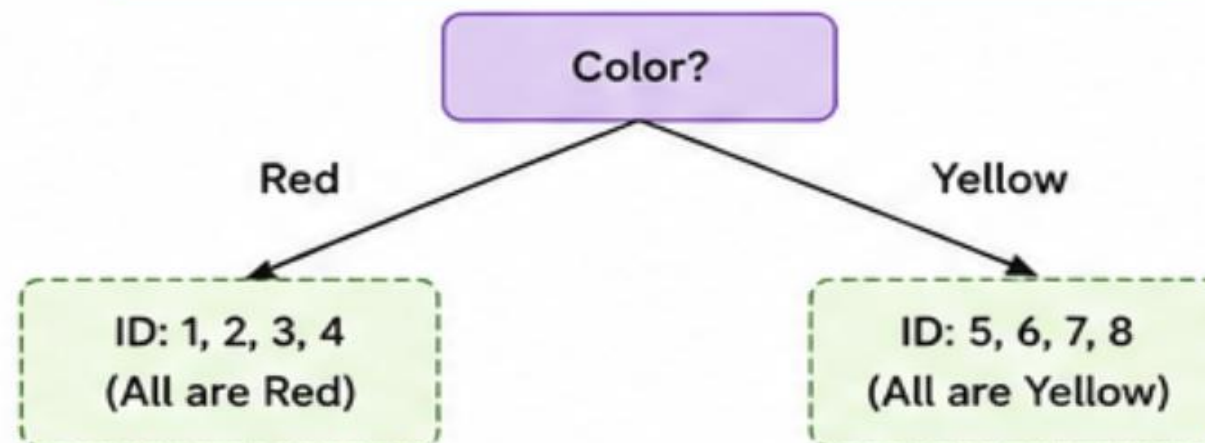
STEP 1

Start with the complete dataset.
Select the best feature as the Root Node.
(Best feature = Color)

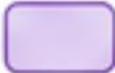


STEP 2

Split the dataset based on Color.
Check each subset.

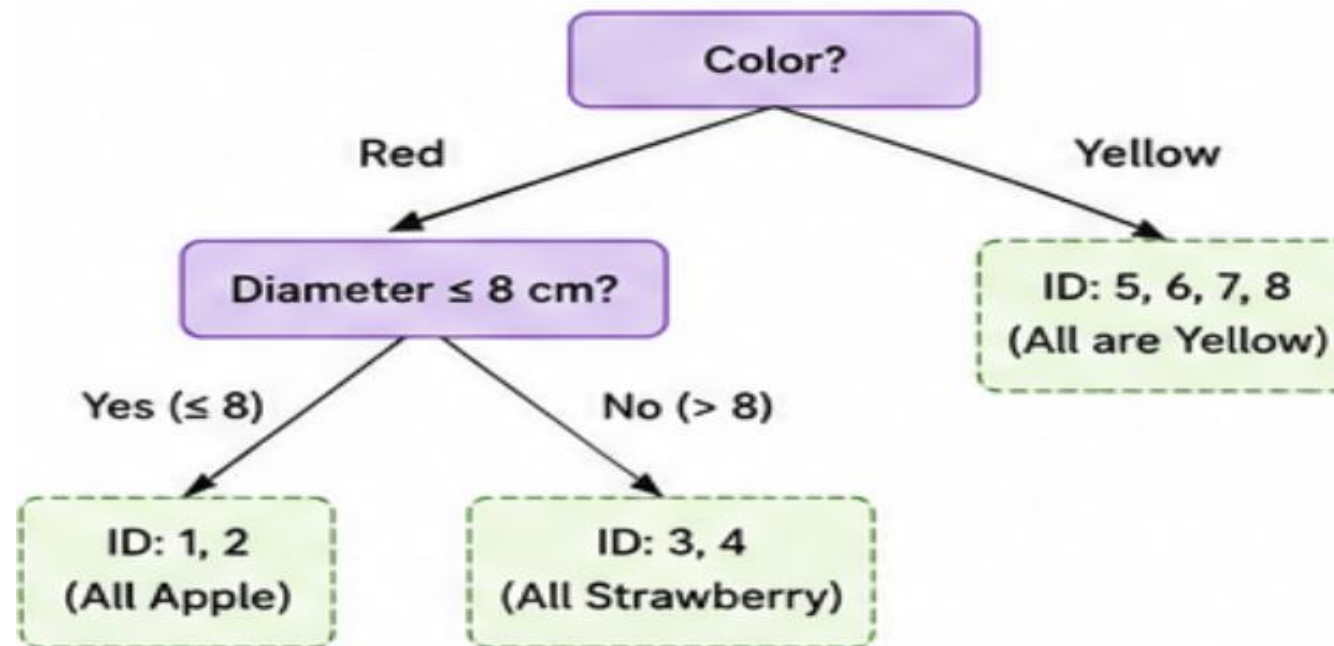


Legend

-  Decision Node (Question)
-  Data Subset
-  Leaf Node (Class Label)

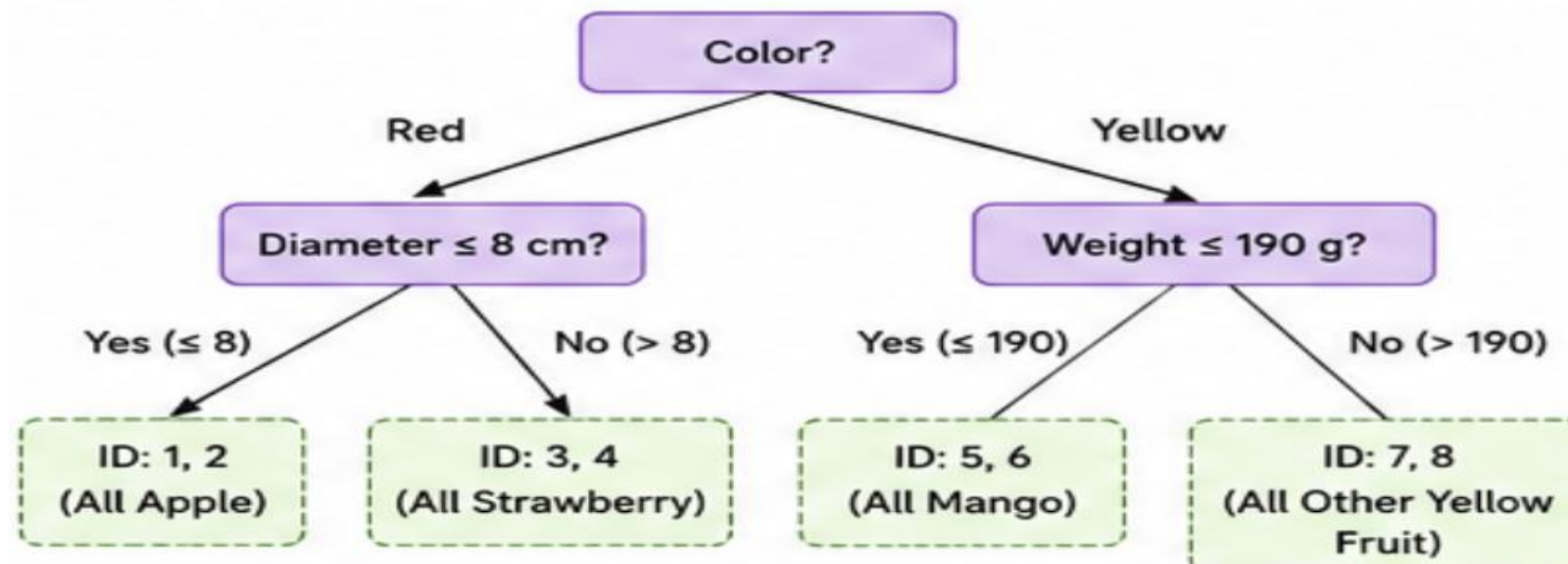
STEP 3

Red subset is not pure (contains Apple and Strawberry). Select next best feature = Diameter and split.



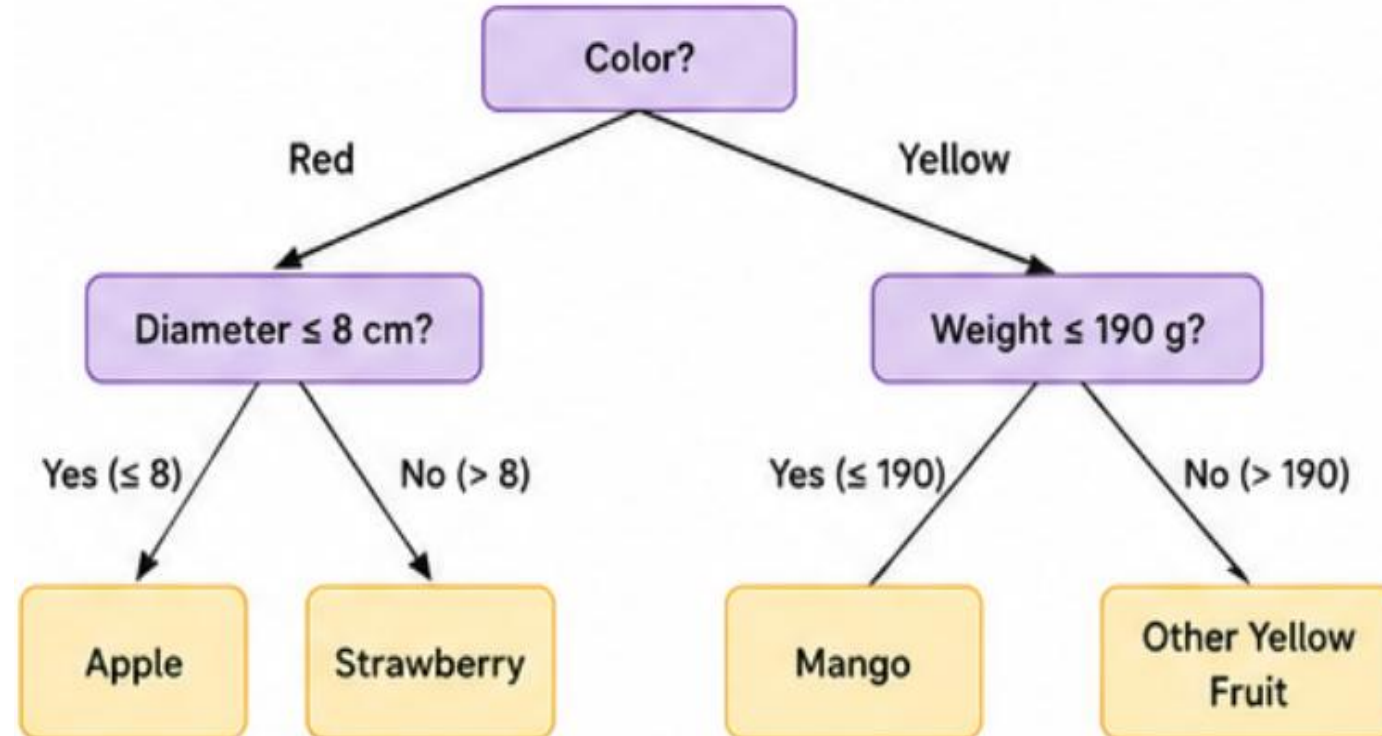
STEP 4

Yellow subset is not pure (contains Mango and Other Yellow Fruit). Select next best feature = Weight and split.



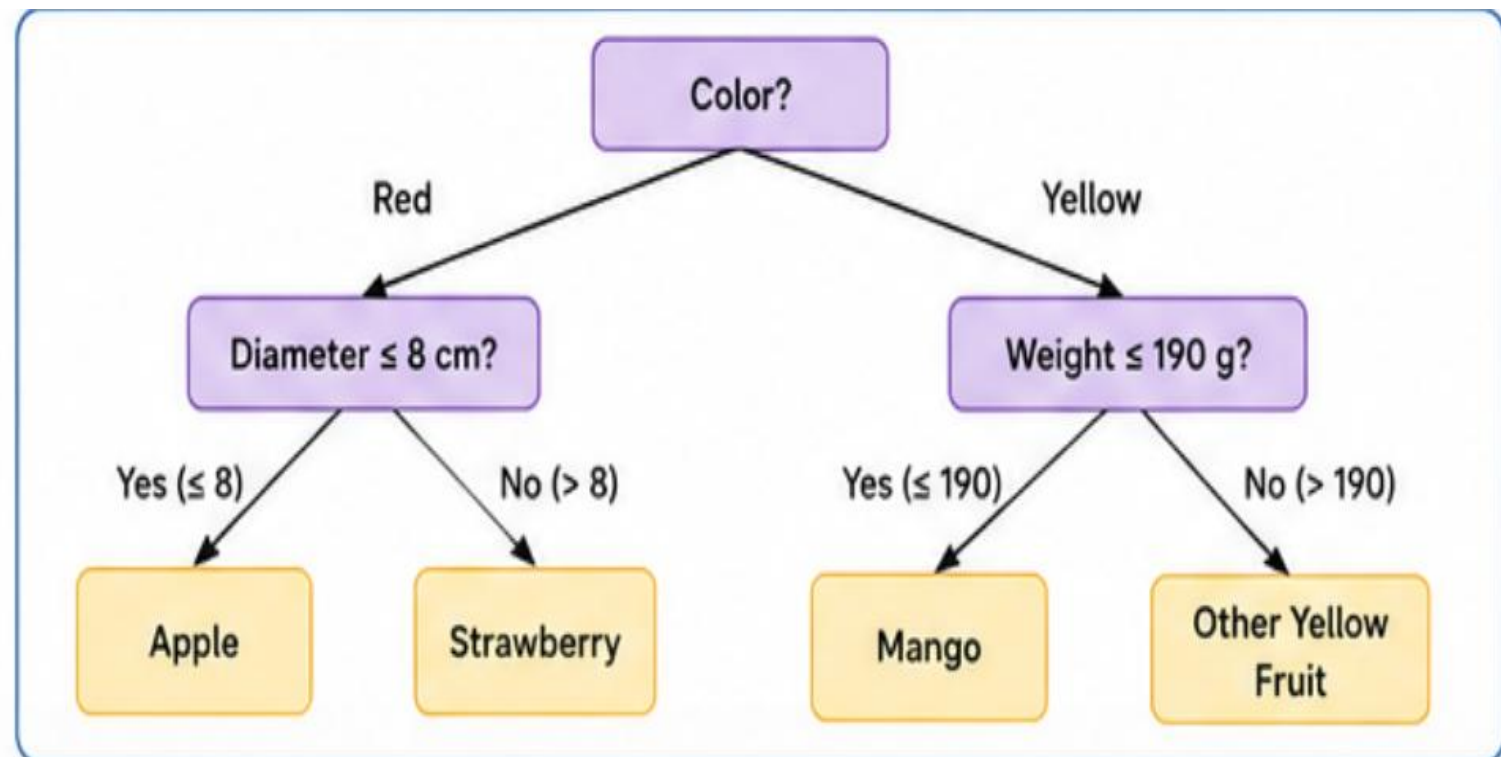
STEP 5

All the subsets are now pure.
Create leaf nodes with their class labels.



STEP 6

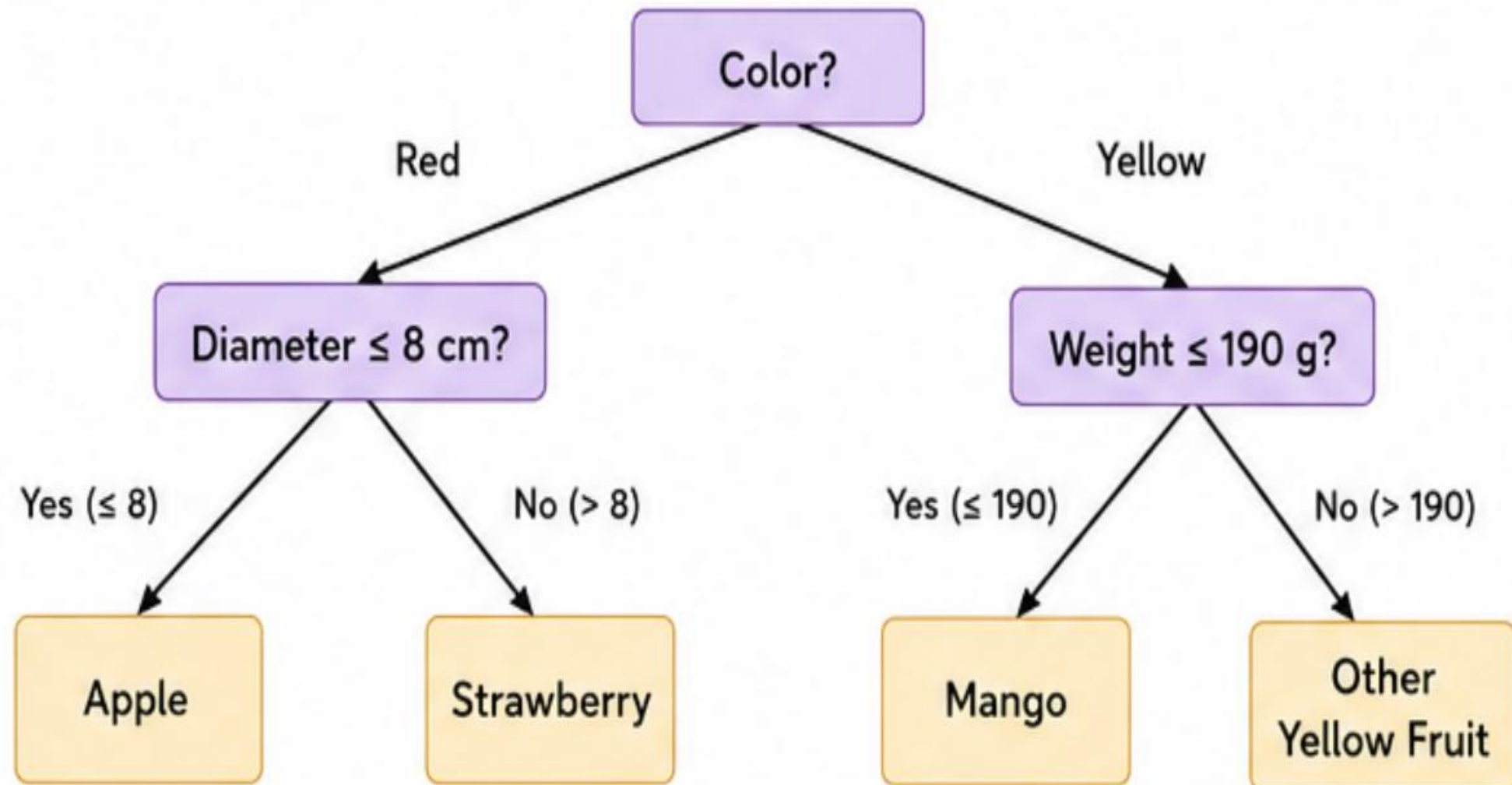
The final Decision Tree is ready and can be used to classify new fruits.



Test Sample

Feature	Value
Weight	165 g
Color	Red
Diameter	8 cm

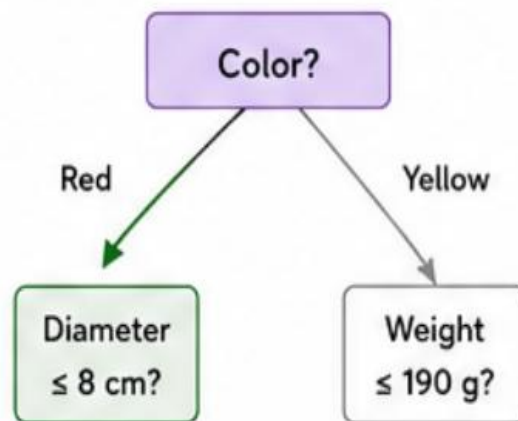
Learned Decision Tree (from Training Data)



How Decision Tree Classifies the Test Sample

Step 1

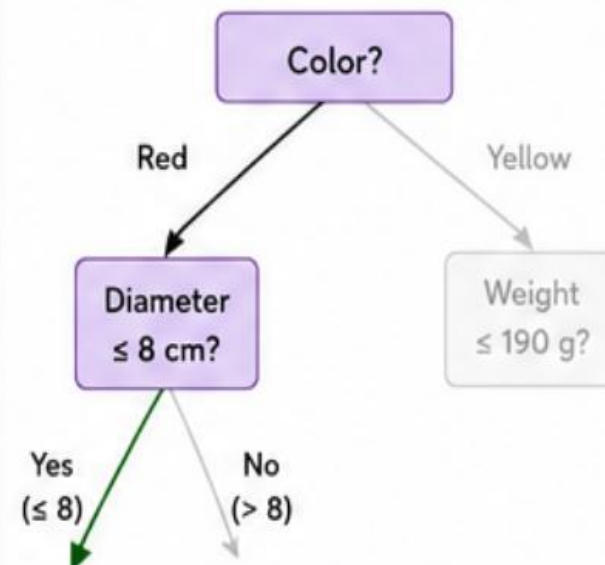
Start at the root node.
Check Color?



Test sample Color = **Red**
So, follow the **Red** branch.

Step 2

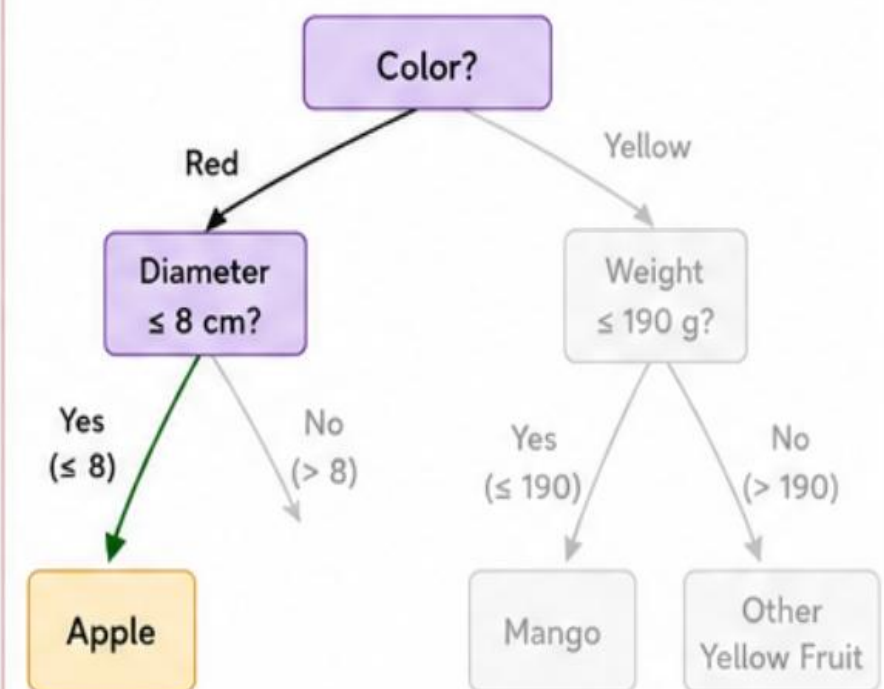
Now at node: Diameter ≤ 8 cm?
Check Diameter.



Test sample Diameter = 8 cm
 $8 \leq 8 \rightarrow$ **Yes**
So, follow the Yes branch.

Step 3

Follow the branch to the leaf node.



We reach the leaf node.

Thus, for the given test sample:

- Weight = 165 g
- Color = Red
- Diameter = 8 cm

The Decision Tree predicts
the fruit as $\rightarrow$ **Apple**

Model Evaluation

In Machine Learning, we first divide the dataset into Training Data and Testing Data. The model learns patterns from the training data. Then we evaluate its performance using the testing data."

Our fruit dataset is very small and simple. The patterns are easy to learn. For example, almost all Red fruits are Apple and Yellow fruits are Mango. Therefore, the Decision Tree correctly classified every test sample, resulting The Decision Tree achieved 100% accuracy

Model Evaluation

Real-world datasets are much more complex. Different classes often have overlapping features, missing values, or noisy data. Therefore, models usually make some incorrect predictions, and the accuracy is often less than 100%."

Here, Color alone cannot distinguish the fruits, so the model has to consider multiple features and may still make mistakes.

Training Data

	Weight	Color	Diameter	Fruit
0	160	Red	8	Apple
1	170	Red	8	Apple
2	180	Yellow	9	Mango
3	220	Yellow	10	Mango
4	165	Red	8	Apple
5	200	Yellow	9	Mango
6	175	Yellow	8	Apple
7	210	Yellow	10	Mango

Testing Data

	Weight	Color	Diameter	Fruit
0	185	Red	9	Mango
1	168	Red	8	Apple
2	205	Yellow	9	Mango
3	190	Yellow	8	Apple

Why do we need Model Evaluation?

Model Evaluation helps us measure how well a machine learning model performs on unseen data. It tells us whether the model's predictions are reliable enough to use in real-world applications.

It helps us answer questions like:

Is the model making correct predictions?

How many mistakes is it making?

Can we trust this model?

Is this model better than another model?

"Without model evaluation, we would not know which algorithm performs better."

There is no single machine learning algorithm that is best for every problem. Different datasets have different characteristics, so different algorithms perform differently.

Model Evaluation?

Model Evaluation is the process of measuring how well a trained machine learning model performs on unseen (test) data.

It helps us determine whether the model makes correct predictions and whether it can be trusted for real-world applications.

Why is Model Evaluation Important?

Measures the model's performance.

Checks how accurately the model predicts new data.

Compares different machine learning algorithms.

Helps select the best model.

Detects overfitting and underfitting.

Advantages

- Simple and Results can be displayed as a tree diagram, making interpretation straightforward.
- Works with both numbers and categories
- Once trained, it quickly predicts the class of new data.

Limitations

- Sensitive to small changes in the dataset..
- it performs very well on the training data but poorly on new, unseen data (Overfitting).
- Performance can be affected by imbalanced datasets
- Large trees become difficult to interpret.

New Fruit

Weight	Color	Fruit
185	Red	Mango

The model predicts:

Red → Apple

But the correct answer is:

Mango

The model **memorized** the training pattern and could not **generalize** to a new case.

This is **Overfitting**.

Underfitting means the model has not learned enough from the training data.

Overfitting means the model has learned too much from the training data, including mistakes or noise.

Evaluation Metric:

Accuracy

Precision

Recall

F1-Score

Confusion Matrix

A Confusion Matrix is a table used to evaluate the performance of a classification model.

It compares the model's predicted labels with the actual labels and shows how many predictions are correct and how many are incorrect.

It tells us where the model is making correct predictions and where it is making mistakes.

Decision Tree classifies fruits. After testing, we obtain:

Test Dataset		
Fruit ID	Actual Fruit	Predicted Fruit
F1	Apple	Apple
F2	Mango	Mango
F3	Apple	Mango
F4	Mango	Mango
F5	Apple	Apple

Can you answer these questions?

- Which fruit was classified incorrectly?
- How many Apples were predicted correctly?
- How many Mangoes were predicted correctly?
- What type of mistakes did the model make?

Confusion Matrix for the Fruit Example

Actual \ Predicted	Apple	Mango
Apple	2	1
Mango	0	2

Interpretation

2 Apples were correctly classified as Apple.

2 Mangoes were correctly classified as Mango.

1 Apple was incorrectly classified as Mango.

No Mango was incorrectly classified as Apple.

Thus, the model made 4 correct predictions and 1 incorrect prediction.

Confusion Matrix for the Fruit Example

Actual \ Predicted	Apple	Mango
Apple	2	1
Mango	0	2

General Confusion Matrix (Binary Classification)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

where one class is designated as the **positive class** and the other as the **negative class**. binary classification

The matrix shows where the model is "confused."

Confusion Matrix

		Predicted Fruit		
		Apple	Mango	Total
Actual Fruit	Apple	2 (TP)	1 (FN)	3
	Mango	0 (FP)	2 (TN)	2
	Total	2	3	<u>5</u>

True Positive (TP): The model correctly predicts the positive class.

True Negative (TN): The model correctly predicts the negative class.

False Positive (FP): The model predicts Positive, but the actual class is Negative.

False Negative (FN): The model predicts Negative, but the actual class is Positive.

1. Accuracy: Accuracy is the percentage of total predictions that are correctly classified by the model.

Or

Accuracy is the proportion of correctly classified instances among all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision: Precision is the proportion of correctly predicted positive samples out of all samples predicted as positive.

Or

Precision measures how many of the instances predicted as positive are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Recall (Sensitivity or True Positive Rate) : Recall is the proportion of actual positive samples that are correctly identified by the model.

Or

Recall measures how many of the actual positive instances are correctly identified by the classifier.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score: F1-Score is the harmonic mean of Precision and Recall. It provides a balanced measure when both Precision and Recall are important.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

or equivalently,

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN}$$

Compare the prediction with the actual value.

Confusion Matrix

		Predicted Fruit		
		Apple	Mango	Total
Actual Fruit	Apple	2 (TP)	1 (FN)	3
	Mango	0 (FP)	2 (TN)	2
	Total	2	3	<u>5</u>

TP (True Positive) = 2

TN (True Negative) = 2

FP (False Positive) = 0

FN (False Negative) = 1

- Accuracy = $(TP + TN) / \text{Total}$
= $(2 + 2) / 5 = 0.80 \rightarrow 80\%$
- Precision (Apple) = $TP / (TP + FP)$
= $2 / (2 + 0) = 1.00 \rightarrow 100\%$
- Recall (Apple) = $TP / (TP + FN)$
= $2 / (2 + 1) = 0.67 \rightarrow 66.67\%$
- F1-Score (Apple) = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
= $2 \times (1.00 \times 0.67) / (1.00 + 0.67) = 0.80 \rightarrow 80\%$

Rows represent Actual classes and Columns represent Predicted classes.
Positive class (for this example) = Apple

Applications:

Medical Diagnosis

Predict diseases based on symptoms such as fever, cough, blood pressure, and test results.

Bank Loan Approval

Decide whether to approve or reject a loan using income, credit score, employment status, and repayment history.

Email Spam Detection

Classify emails as Spam or Not Spam using keywords, sender information, and attachments.

Agriculture

Recommend suitable crops based on soil type, rainfall, temperature, and season.

Quality Control

Detect defective and non-defective products in manufacturing using measurements and inspection results.

Exercise

A machine learning model was developed to predict whether an email is Spam (Positive) or Not Spam (Negative).

When tested on 100 emails, the results were as follows:

- The model correctly predicted 40 spam emails as spam
- It correctly predicted 50 non-spam emails as not spam
- It incorrectly marked 5 non-spam emails as spam
- It failed to detect 5 spam emails and marked them as not spam
- Construct Confusion matrix and calculate accuracy, precision, recall.

Training

The system is given thousands of emails.

Email

"Win ₹1 Crore!"

"Meeting at 2 PM"

"Claim your prize"

"Project Report"

Label

Spam

Not Spam

Spam

Not Spam

The machine learns patterns:

- "Prize"
- "Free"
- "Lottery"

are common in spam emails.

Testing

New Email:

"Congratulations! Claim your free reward."

The machine compares it with what it learned during training and predicts:

➡ **Spam**



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ML Supervised Learning

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K-Nearest Neighbors (KNN) Algorithm

Definition

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression problems. It classifies a new data instance by finding the K closest training examples in the feature space and assigning the class that appears most frequently among them.

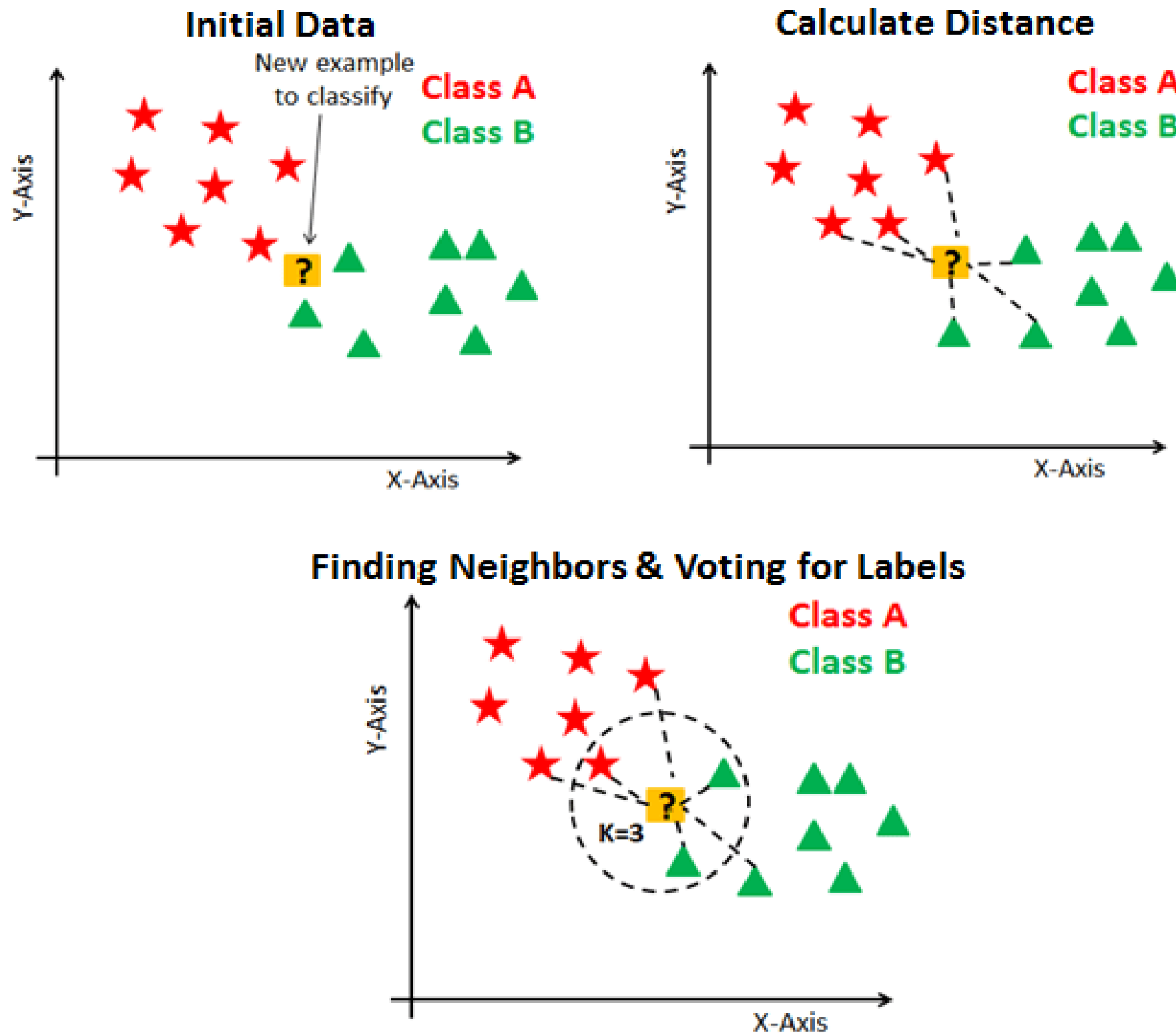
KNN is called a **Lazy Learning Algorithm** because it does not build a model during the training phase. Instead, it stores all the training data and performs computation only when a new data point needs to be classified.

Principle of KNN

- When a new data point is received,
- KNN: Measures the distance between the new point and every training instance.
- Finds the K nearest neighbors.
- Predicts the class based on majority voting.

K is the number of nearest neighbors that the algorithm considers before making a prediction.

Working of KNN



Importance of Selecting the Value of K in KNN

The value of K in the K-Nearest Neighbors (KNN) algorithm represents the number of nearest neighbors considered when predicting the class of a new data point.

Selecting an appropriate value of K is important because it directly affects the accuracy, stability, and performance of the KNN model.

Small K (e.g., $K = 1$)

When $K = 1$, the algorithm considers only the nearest neighbor. May produce incorrect predictions if the nearest neighbor is mislabeled.

Moderate K (e.g., $K = 3$ or $K = 5$)

The algorithm considers multiple neighbors and predicts using majority voting.

Produces more reliable predictions. Usually gives better generalization.

Algorithm

Step 1: Store all training data.

Step 2: Choose the value of K.

Step 3: Calculate the distance between the new data point and every training instance.

The most commonly used distance is Euclidean Distance.

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

where

x_i = feature value of test sample

y_i = feature value of training sample

n = number of features

Step 4: Sort all distances in ascending order.

Step 5: Select the first K nearest neighbors.

Step 6: Count the number of occurrences of each class among the K neighbors.

Step 7: Assign the class with the highest frequency to the new data point.

Step 8: Display the predicted class.

1. TRAINING DATASET

Patient	Glucose (mg/dL)	BMI (kg/m ²)	Class
P1	80	22	No Diabetes
P2	85	24	No Diabetes
P3	90	26	No Diabetes
P4	140	32	Diabetes
P5	150	34	Diabetes
P6	160	36	Diabetes

 **No Diabetes** **Diabetes**

NEW PATIENT (TO BE CLASSIFIED)

Glucose = 145 mg/dL

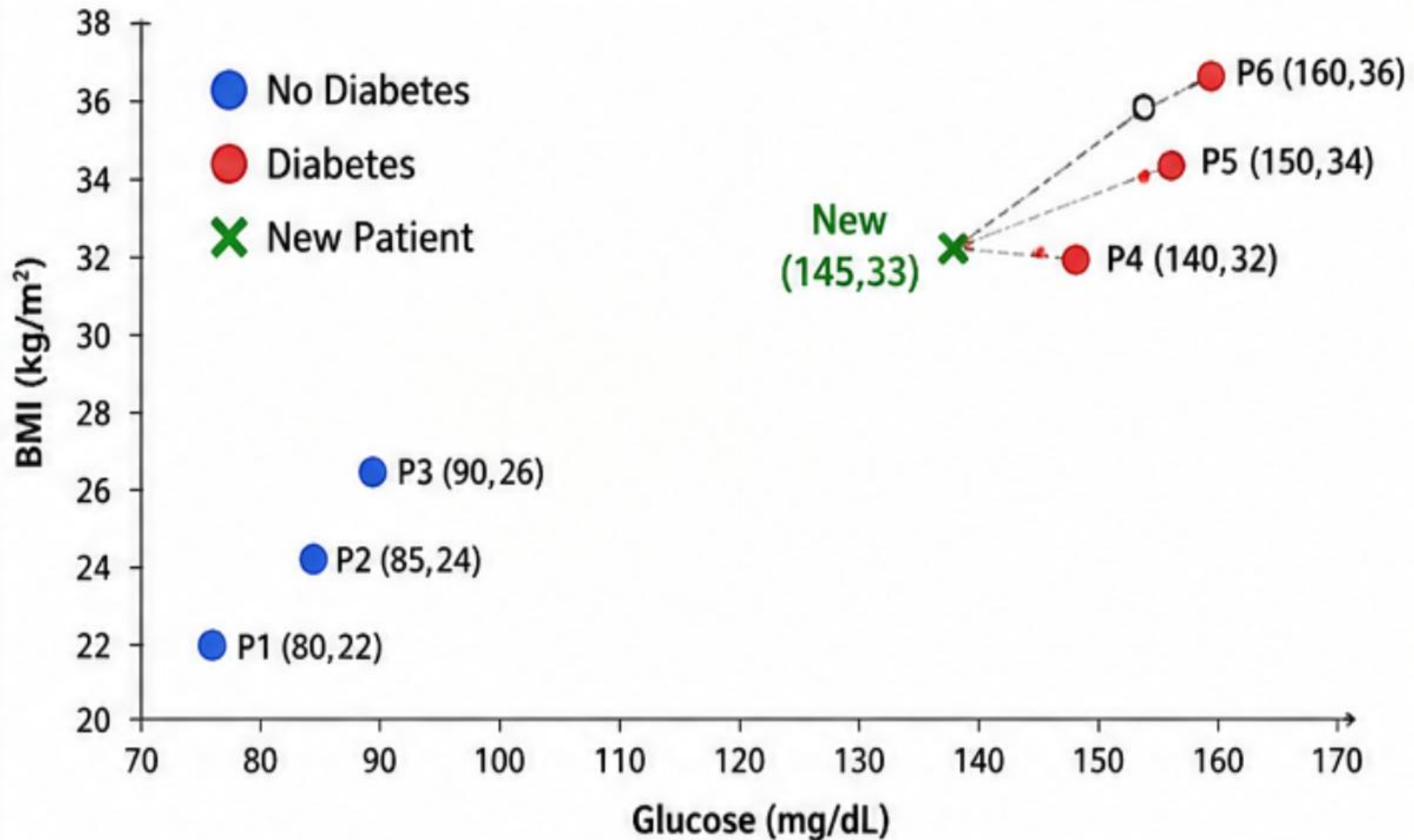
BMI = 33 kg/m²

✕ (145, 33)

Choose **K = 3**

The algorithm considers the three nearest neighbors.

2. DATA POINTS IN 2D SPACE



3. CALCULATE EUCLIDEAN DISTANCE

Distance formula:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

where

$$x_1 = 145, \quad x_2 = 33$$

Patient	Calculation	Distance
P1	$\sqrt{(145-80)^2 + (33-22)^2}$	= 65.92
P2	$\sqrt{(145-85)^2 + (33-24)^2}$	= 60.67
P3	$\sqrt{(145-90)^2 + (33-26)^2}$	= 55.44
P4	$\sqrt{(145-140)^2 + (33-32)^2}$	= 5.10
P5	$\sqrt{(145-150)^2 + (33-34)^2}$	= 5.10
P6	$\sqrt{(145-160)^2 + (33-36)^2}$	= 15.30

Here:

- $x_1 = 145 \rightarrow$ Feature 1 of the test sample
- $x_2 = 33 \rightarrow$ Feature 2 of the test sample

Test patient:

$$(x_1, x_2) = (145, 33)$$

Training patient P1:

$$(y_1, y_2) = (80, 22)$$

Euclidean Distance

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Substitute the values:

$$d = \sqrt{(145 - 80)^2 + (33 - 22)^2}$$

$$d = \sqrt{65^2 + 11^2}$$

$$d = \sqrt{4225 + 121}$$

$$d = \sqrt{4346}$$

$$\boxed{d = 65.92}$$

4. SORT DISTANCES (ASCENDING ORDER)

Rank	Patient	Distance	Class
1	P4	5.10	Diabetes
2	P5	5.10	Diabetes
3	P6	15.30	Diabetes
4	P3	55.44	No Diabetes
5	P2	60.67	No Diabetes
6	P1	65.92	No Diabetes

5. SELECT $K = 3$ NEAREST NEIGHBORS

Selected Neighbors ($K=3$)	Class
P4 (Distance = 5.10)	Diabetes
P5 (Distance = 5.10)	Diabetes
P6 (Distance = 15.30)	Diabetes

6. MAJORITY VOTING

Diabetes = 3 votes No Diabetes = 0 votes

7. PREDICTION

The new patient is predicted to have **DIABETES**

Pros

- Very simple and easy to understand and implement.
- Works very well for datasets with less number of features
- KNN can be used to solve both classification (e.g., spam detection) and regression (predicting continuous values) problems.
- KNN performs well when the dataset is small and the classes are well separated.
- Few parameters: k , Distance Metric

Cons

- To classify a new sample, KNN calculates the distance to all training samples. This becomes slow for large datasets.
- KNN stores the entire training dataset, which requires more memory.
- Incorrect or noisy training samples can influence the prediction, especially when K is small.
- As the number of features increases, distances between data points become less meaningful. This is known as the curse of dimensionality.

Applications :

- KNN is widely used in applications where similarity between data points is important. Since KNN predicts a new instance based on its nearest neighbors, it is suitable for recommendation, classification, and pattern recognition tasks.
- 1. **Image Classification:** A new image of an animal is compared with stored images. If most similar images are cats, the new image is classified as a cat.
 - Animal identification
 - Plant species classification
 - Object recognition
- **2. Face Recognition**
 - Smartphone face unlock
 - Attendance systems
 - Security and surveillance

Applications :

- **3. Email Spam Detection**

- Classify emails as Spam or Not Spam. A new email is compared with previously labeled emails. If most similar emails are spam, it is marked as spam.

- **4. Recommendation Systems**

- Recommend products, movies, or songs based on user preferences.
- Movie recommendations
- Music recommendations
- Online shopping

- **5. Customer Segmentation:** Group customers with similar purchasing behavior.

- **6. Document and Text Classification :**

news article is classified as:

Sports, Politics, Technology, Entertainment

by comparing it with previously labeled articles.

Practice

Name	Acid Durability	Strength	class
Type-1	7	7	Bad
Type-2	7	4	Bad
Type-3	3	4	Good
Type-4	1	4	Good
Type-5	3	7	??



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ML Supervised Learning

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Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression problems.

It works by finding the optimal hyperplane that separates different classes with the maximum possible margin.

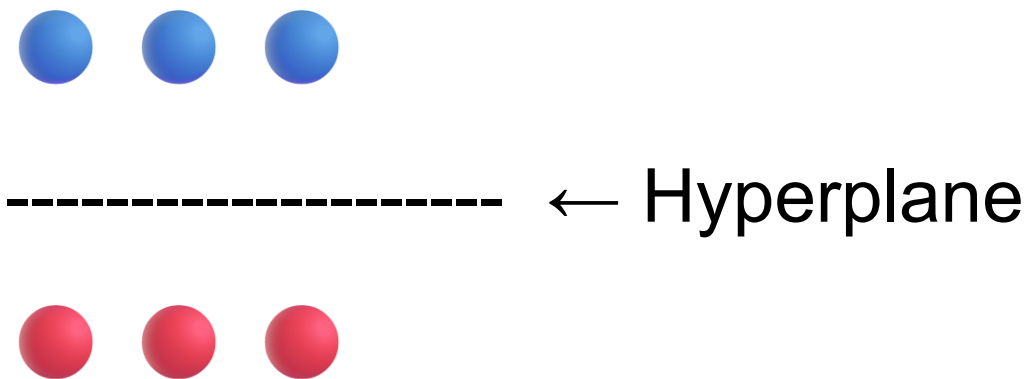
The main **objective of SVM** is to maximize the distance (margin) between the separating hyperplane and the nearest data points from each class, resulting in better generalization and classification accuracy.

Tom M. Mitchell

"A Support Vector Machine is a learning algorithm that constructs a hyperplane or set of hyperplanes in a high-dimensional space for classification or regression."

- Imagine you have two groups of students:
 - Some prefer Cricket
 - Some prefer Football
- You want to separate them in a playground.
- You draw a line such that:
 - Cricket lovers are on one side
 - Football lovers are on the other side
- This is what SVM does – it finds the best dividing line (hyperplane)

- Works by **finding a hyperplane that best separates classes**
- Goal: **Maximize the margin** between the classes and the dividing line
- Support vectors = Data points closest to the line



Margin

Margin means empty space around the hyperplane. The blank space above and below the line is called the Margin.

SVM always tries to make this margin as wide as possible.

A larger margin means:

Better separation

Less confusion

Better prediction on new data

Less overfitting

That is why SVM looks for the maximum margin hyperplane.

Small Road



Vehicles may collide.

Wide Road



Safer.

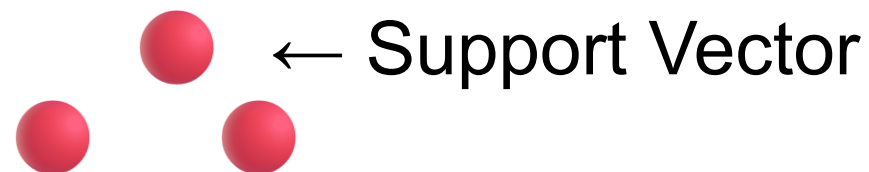
- **Hyperplane** → Best separating boundary
- **Margin** → Distance between hyperplane & closest data points
- **Support Vectors** → Points that lie on margin boundaries and influence the hyperplane

Working of SVM

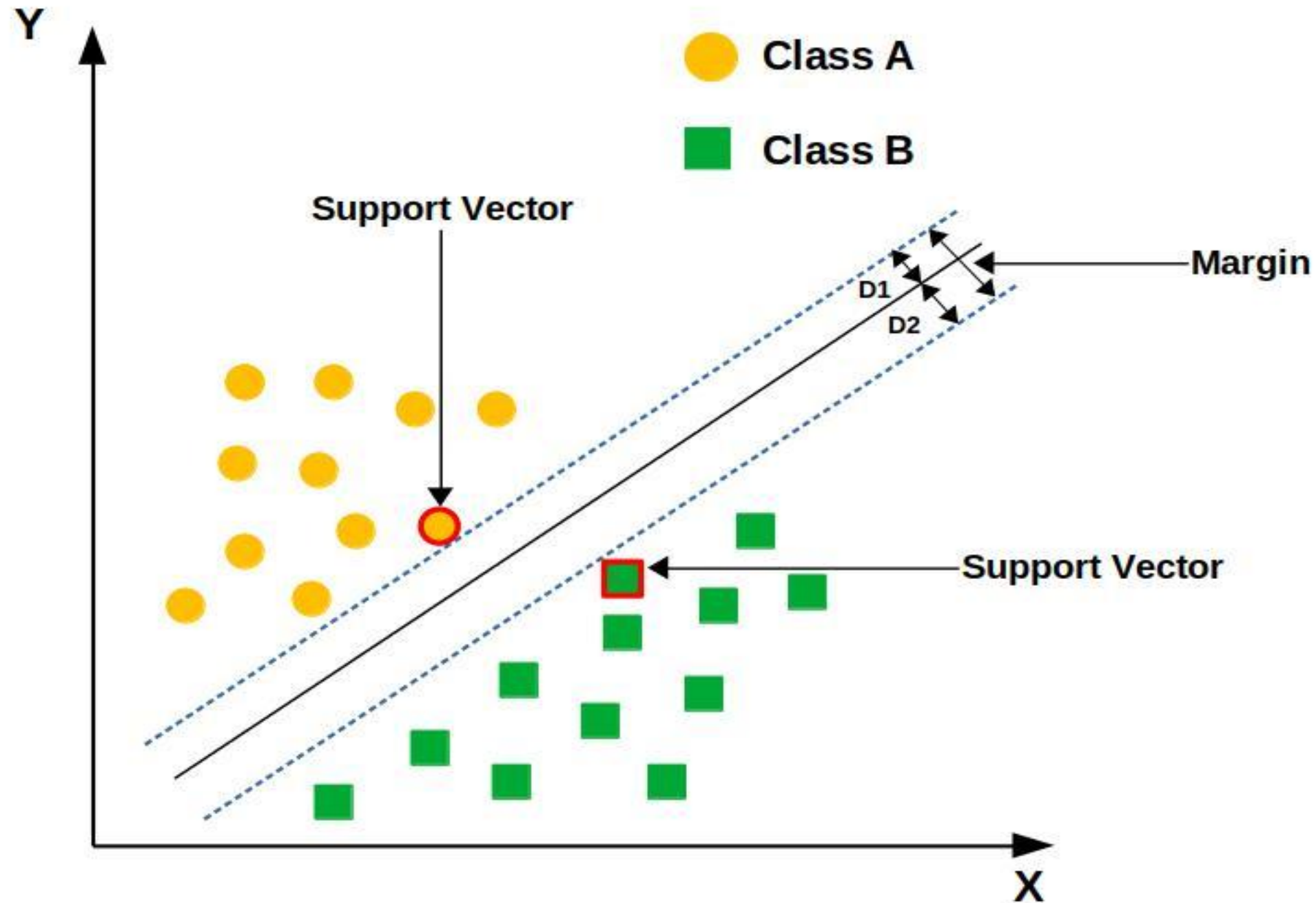
1. Plot data points of two classes
2. Find a line (hyperplane) that separates the classes
3. Choose the one with the maximum margin (largest distance between nearest points of classes)
4. Use this hyperplane to classify new data



=====



the points nearest to the hyperplane called Support Vectors. The other points far away don't affect the hyperplane



The SVM algorithm:

1. Looks at all training points.
2. Finds many possible separating lines.
3. Calculates the margin for each line.
4. Chooses the line with the **maximum margin**.
5. Identifies the **support vectors** (closest points).
6. Stores the decision boundary as the trained model.

SVM find the (shortest) distance from each data point to the hyperplane.

What is the hyperplane?

With one feature:

Hyperplane is point/threshold

With 2 features:

The hyperplane is a **line**. Represent in 2D

With three feature:

the data exists in **3-dimensional space**. SVM decision boundary becomes a **plane**.

More than three feature means Hyperplane.

Support Vectors

The distance between the boundary and the closest points is the **margin**.

Applying SVM to the following dataset

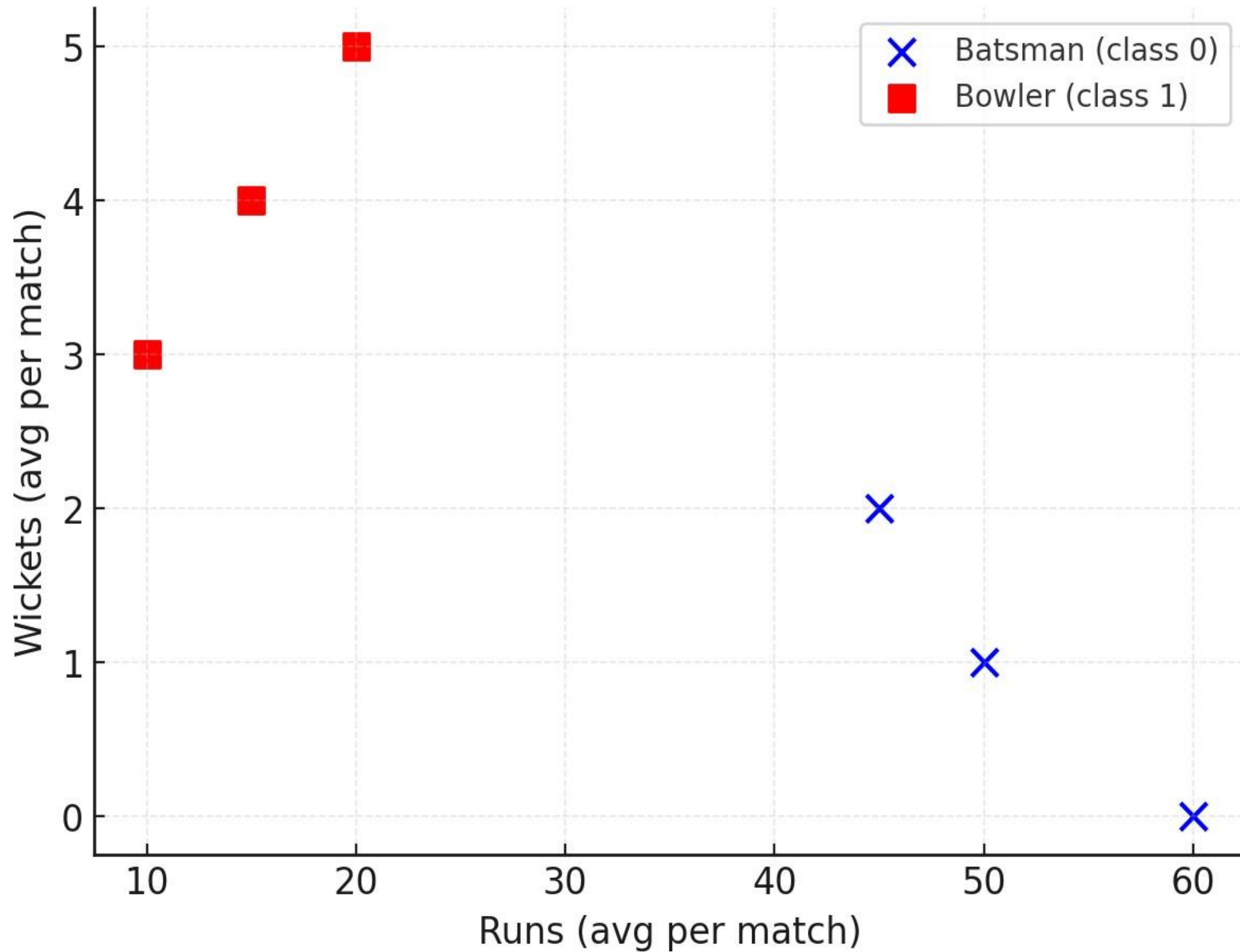
Player	Avg Runs	Avg. Wickets	Class
P1	50	1	Batsman
P2	60	0	Batsman
P3	45	2	Batsman
P4	10	3	Bowler
P5	15	4	Bowler
P6	20	5	Bowler

1. How machine will train. The machine starts looking for relationships.
2. Avg runs and Avg Wickets are the two features.

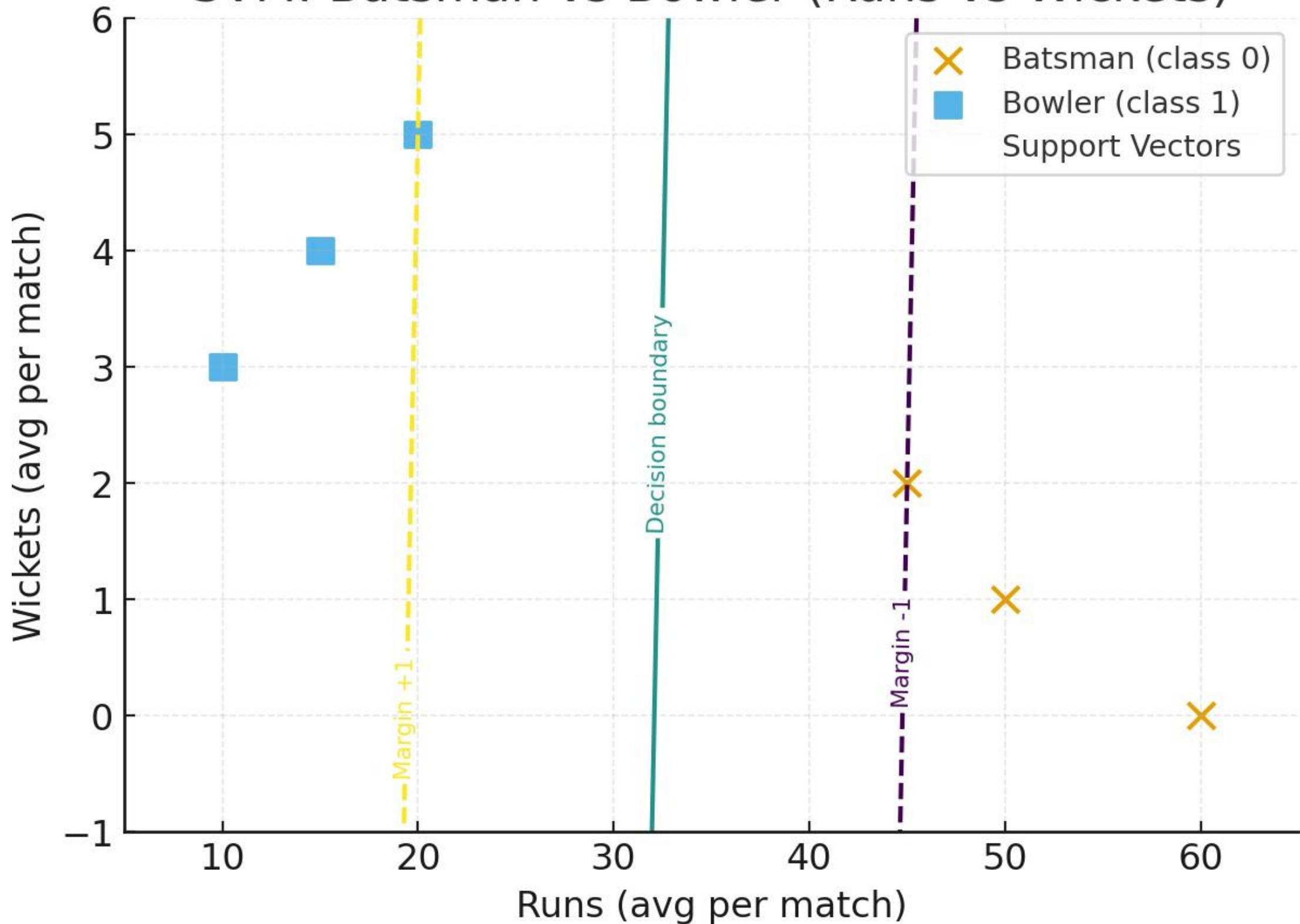
1. Low Runs, High Wickets Mostly Bowlers
2. High Runs, Low Wickets Mostly Batsmen

1. For SVM, the machine finds the best hyperplane that separates the two classes.
2. After training, the machine stores the decision boundary.
3. Suppose a new player comes.
4. The model checks: "Which side of the hyperplane is this player?"

Cricket Dataset: Batsman vs Bowler



SVM: Batsman vs Bowler (Runs vs Wickets)



The two points **closest to the other class** are:

- Bowler → **P6 = (20, 5)**
- Batsman → **P3 = (45, 2)**

These are the **support vectors**

using a **simplified SVM where Runs alone determines the separation**, the decision boundary is exactly halfway between their Run values:

$$\begin{aligned}\text{Decision Boundary} &= \frac{\text{Bowler support vector} + \text{Batsman support vector}}{2} \\ &= \frac{20 + 45}{2} \\ &= \boxed{32.5}\end{aligned}$$

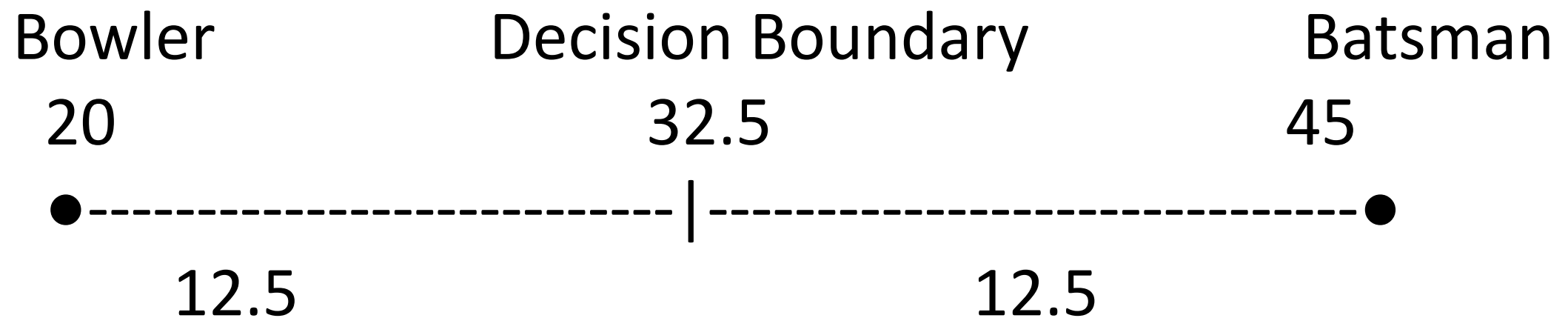
So the decision boundary is:

$$\boxed{x = 32.5}$$

The boundary is equally far from both support vectors:

$$32.5 - 20 = 12.5$$

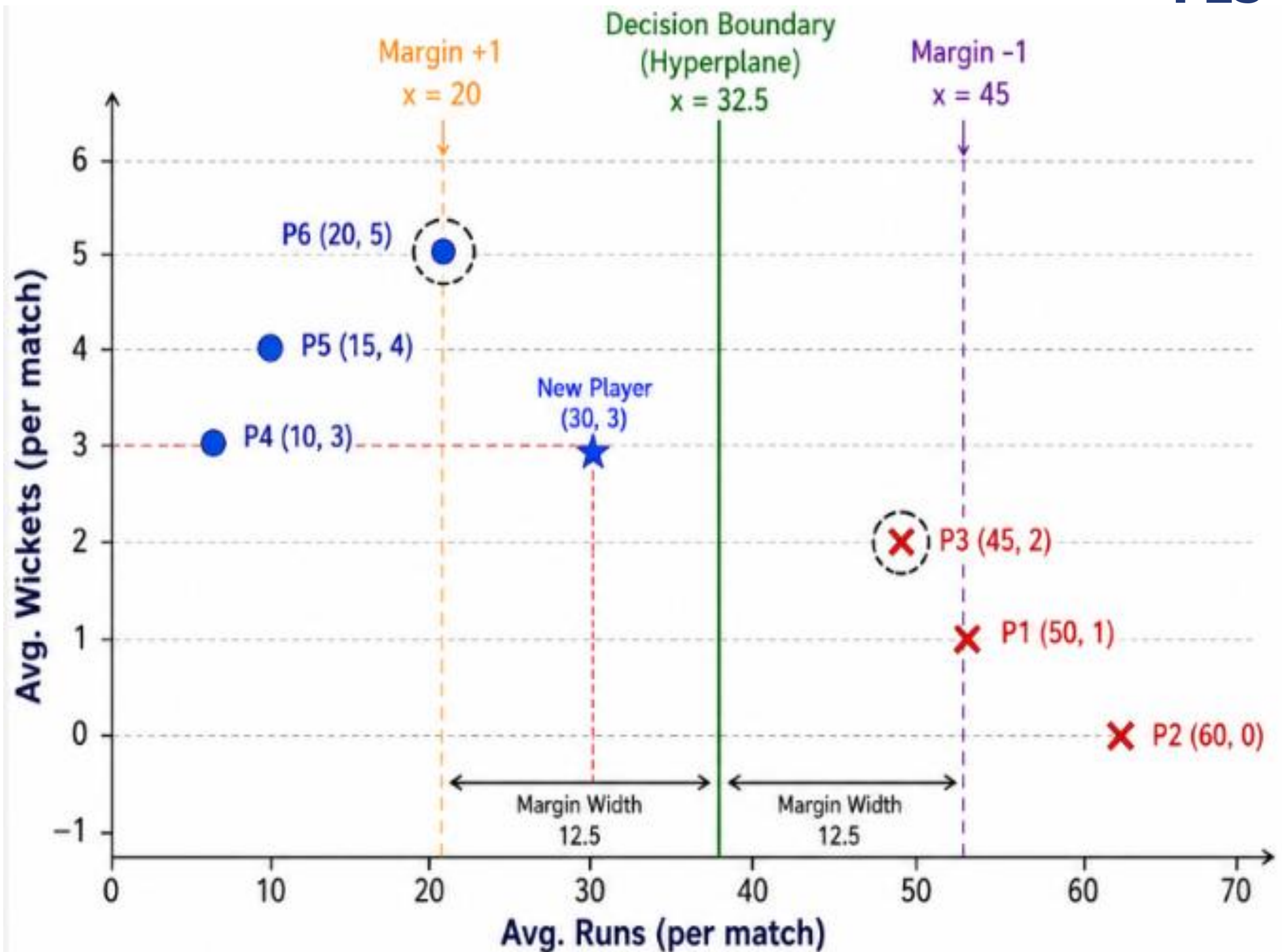
$$45 - 32.5 = 12.5$$



Here consider the only one feature : runs.

- Line roughly between the two groups:
 - Above line → Bowler
 - Below line → Batsman
- Identify the support vectors → Closest points from each class that “support” the margin:
 - Batsman near the line (eg, P3)
 - Bowler near the line (eg, P4)

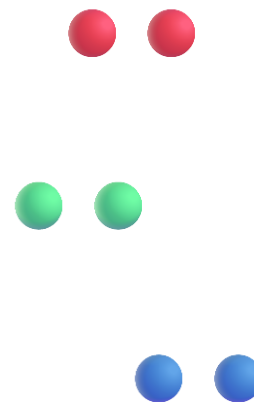
- Suppose a new player has: **Runs = 30, Wickets = 3**
- Check which side of the line the player lies:
 - If above → Bowler
 - If below → Batsman”



What if there are 3 classes?

Suppose we have:

- Apple
- Mango
- Orange



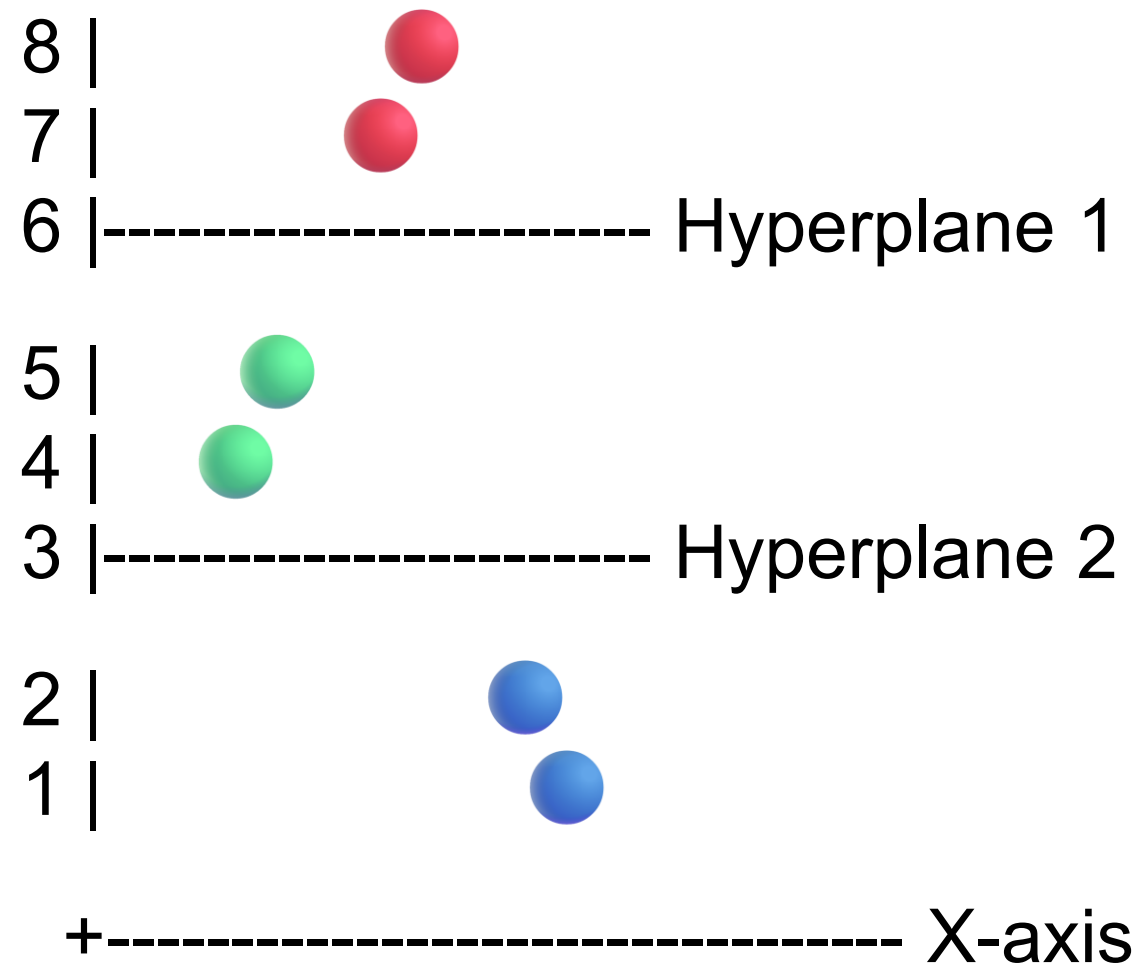
Can one hyperplane separate all three classes?

SVM is a binary classifier.

That means it can classify only 2 classes at a time.

SVM creates multiple hyperplanes.

Y-axis



- Apple
- Mango
- Orange

Take only two classes at a time and ignore all the others.

One-vs-One

Classifier 1

Apple vs Mango

Classifier 2

Apple vs Orange

Classifier 3

Mango vs Orange

Suppose we build **Apple vs Rest.** (One VS rest)

Pros

- Works well with high-dimensional data
- Effective when there is a clear margin of separation
- Good performance for small/medium datasets
- The maximum margin principle helps SVM generalize well to unseen data, reducing overfitting.
- SVM often provides excellent performance even when the training dataset is not very large.

Cons

- Computationally expensive for large datasets
- Poor performance when data is noisy/overlapping
- Less interpretable:
- Unlike Decision Trees, the decision-making process of SVM is not easy to understand or visualize.
- Training time increases as the dataset size and feature count grow.

Applications

Write the real time application where SVM algorithm is suitable

Student	Study Hours (X_1)	Attendance (X_2)	Class
S1	1	50	Fail
S2	2	55	Fail
S3	2	60	Fail
S4	3	65	Pass
S5	4	70	Pass
S6	5	75	Pass

Classify a student

Suppose a new student has:

- Study Hours = 4
- Attendance = 70%



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Regression analysis is a statistical method for modeling and analyzing the relationship between a dependent variable and one or more independent variables.

Regression is used to:

Predict future values. Understand the relationship between variables.

Terminology:

Dependent Variable (Y): Variable to be predicted

Independent Variable (X): Variable used for prediction

Regression Line: Best-fit line showing the relationship

Coefficient: Indicates how much Y changes when X changes

Types of regression:

1. Simple Linear Regression
2. Multiple Linear Regression
3. Polynomial Regression

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value by establishing a linear relationship between the input features (independent variables) and the output (dependent variable).

by fitting the best possible straight line to the observed data.

linear because the model assumes that the relationship between the input variable(s) and the output variable can be **represented by a straight line.**

Examples:

Student marks

House prices

Salary

Temperature

data about students' study hours and marks. Can we predict the marks of a new student?"

Student	Hours Studied	Marks
A	1	40
B	2	50
C	3	60
D	4	70
E	5	80

If a new student studies for 6 hours, how many marks do you think they will get?

observed a pattern.

1 hour → 40 marks

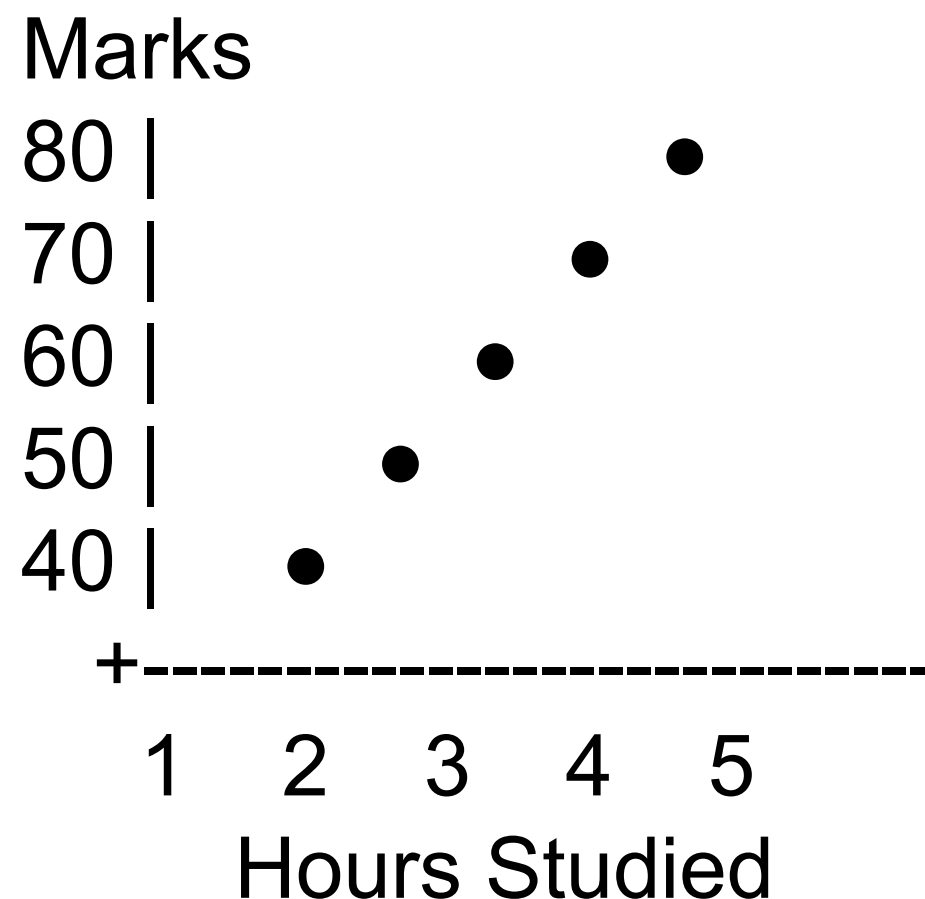
2 hours → 50 marks

.

.

Every extra hour increases the marks by 10.

Machine learns this pattern from the training data and predicts the marks for a new student.



Linear Regression equation:

$$Y = mX + c$$

X = Hours studied (input). This is the input given to the model.

Y = Predicted marks (output) : This is the value we want to predict of a new student.

M (slope) = How much Y changes when X increases by one unit.(Increase in marks for every additional hour)

C (Intercept) = the starting value, i.e., the predicted marks when X=0.

Every additional hour increases the marks by 10.

$$m=10$$

Find the Starting Value (c)

$$Y=mX+c$$

first student studied 1 hour and scored 40 marks.

$$40 = 10(1) + c$$

$$c = 30$$

$$Y = 10X + 30$$
 final regression equation for the given dataset

What is the initial value of Y when student studies 0 hours Substitute X
= 0 (initial hour)

$$Y = 10(0) + 30$$

$$Y = 30$$
 (not required this)

How can we predict the marks for new students, if he study 6 hours?

We use this equation of the line.

$$Y=10X+30$$

Substitute

$$X = 6$$

$$Y=10(6)+30$$

$$Y=90$$

Marks

90 |

80 |

70 |

60 |

50 |

40 |

+

1

2

3

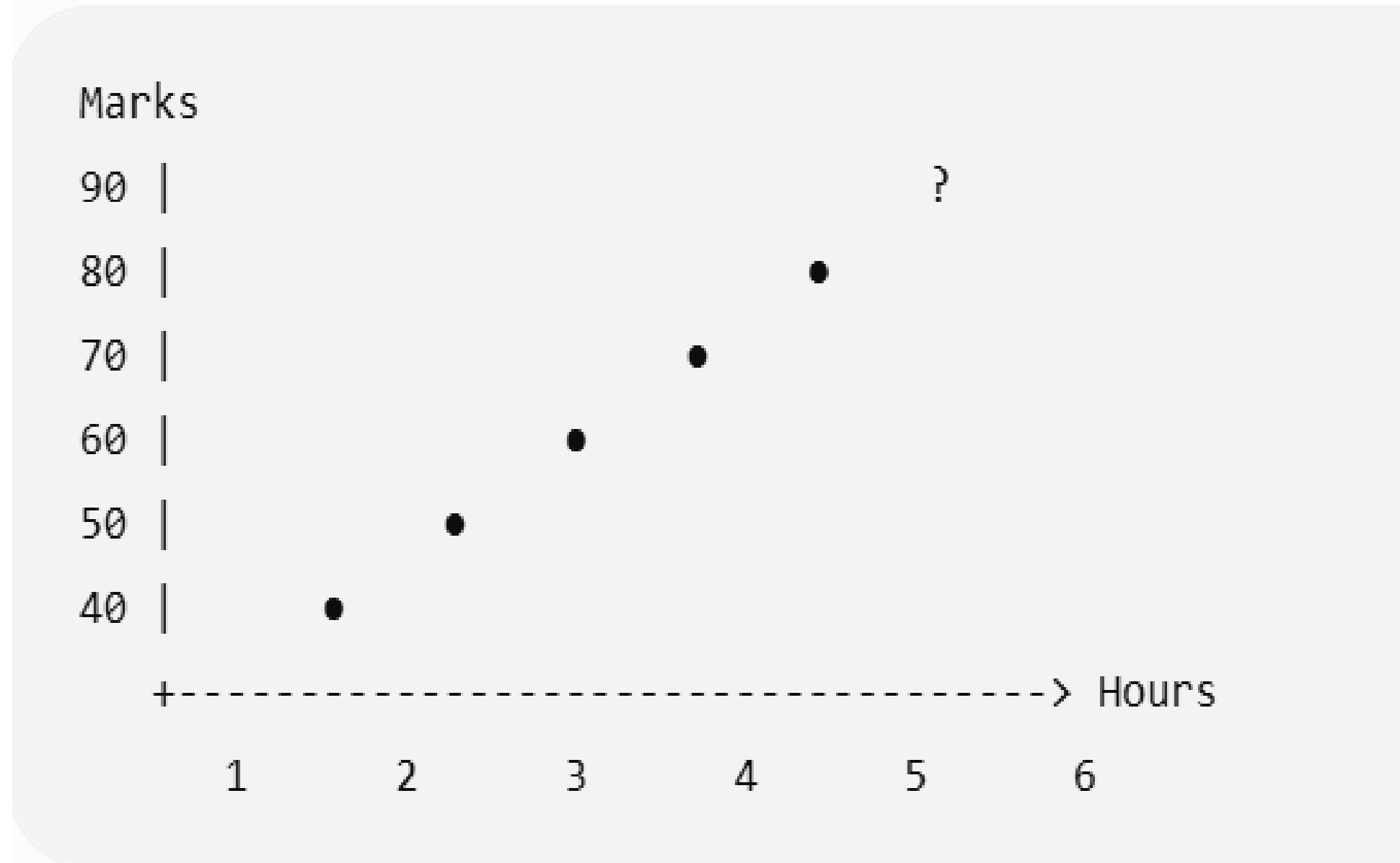
4

5

6

?

Hours



The straight line acts like a rule.

$$Y=10X+30$$

Now it can predict any value.

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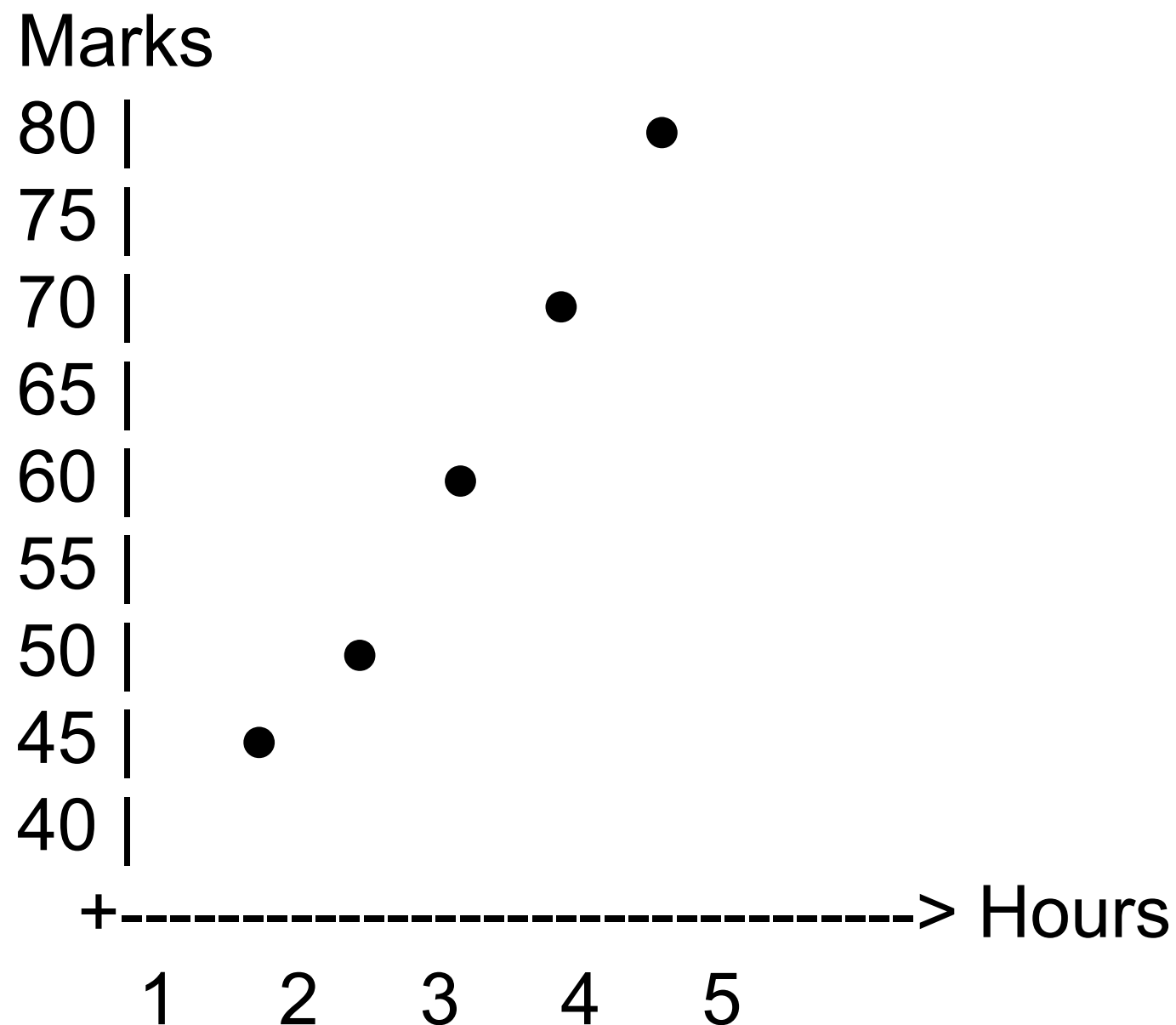
Linear Regression



Hours	Marks
1	42
2	47
3	61
4	69
5	78

What if the Data Is Not Perfect?

Real-world data is not perfect.
The points are not perfectly aligned



In real-world data, all points do not lie exactly on a straight line. Therefore, Linear Regression finds the line that is closest to all the data points.

This is called the best-fit line.

How does the machine find the best-fit line?

It uses the **Least Squares Method**.

Least Squares formulas mathematically calculate the best values of

m

c

instead of guessing them.

This method calculates the prediction error for the line and selects the line with the minimum Sum of Squared Errors (SSE)."

Using these two formulas.

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

Step 2: Calculate the Mean

Mean of X

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$

Mean of Y

$$\bar{y} = \frac{42 + 47 + 61 + 69 + 78}{5} = \frac{297}{5} = 59.4$$

Step 3: Prepare the Table

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	42	-2	-17.4	34.8	4
2	47	-1	-12.4	12.4	1
3	61	0	1.6	0	0
4	69	1	9.6	9.6	1
5	78	2	18.6	37.2	4

Now calculate the totals.

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 34.8 + 12.4 + 0 + 9.6 + 37.2 = 94$$

$$\sum (x_i - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

Step 4: Calculate the Slope (m)

Formula:

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Substitute the values:

$$m = \frac{94}{10} = 9.4$$

So,

$$m = 9.4$$

Step 5: Calculate the Intercept (c)

Formula:

$$c = \bar{y} - m\bar{x}$$

Substitute the values:

$$c = 59.4 - (9.4 \times 3)$$

$$c = 59.4 - 28.2$$

$$c = 31.2$$

So,

$$c = 31.2$$

Step 6: Final Linear Regression Equation

Substitute m and c into

$$Y = mX + c$$

$$Y = 9.4X + 31.2$$

Step 7: Verify the Equation

For 3 hours:

$$Y = 9.4(3) + 31.2$$

$$Y = 28.2 + 31.2$$

$$Y = 59.4$$

Actual marks = **61**

Predicted marks = **59.4**

The prediction is close to the actual value.

This equation is then used to predict the marks of new students based on their study hours."

$$Y=9.4X+31.2$$

For 6 hours:

$$Y = 9.4(6) + 31.2$$

$$Y = 56.4 + 31.2$$

$$Y = 87.6$$

Predicted marks $\approx$ 88

Pros

- Simple & easy to understand
- Fast to train and implement
- Works well for linear relationships
- Useful for prediction and forecasting
- Easy interpretability of coefficients

Cons

- Computationally expensive for large datasets
- Poor performance when data is noisy/overlapping
- Less interpretable

Dataset 1: Salary Prediction (Recommended)

Problem: Predict an employee's salary based on years of experience.

Years of Experience (X)	Salary (₹ in Lakhs) (Y)
1	3
2	5
3	7
4	9
5	11

Dataset 2: House Price Prediction

House Area (sq.ft.)	Price (₹ Lakhs)
600	28
800	36
1000	47
1200	54

Evaluation Metrics in Linear Regression

After training a Linear Regression model, we obtain a regression equation such as

$$Y = mX + c$$

This equation is used to predict the output for new data.

the predicted values may not exactly match the actual values because real-world data contains noise and variations.

it is necessary to evaluate how well the regression model performs.

Evaluation Metrics in Linear Regression

To evaluate the model, we compare the actual values with the predicted values and calculate various evaluation metrics.

The commonly used evaluation metrics are:

1. Residual (Error)
2. Mean Squared Error (MSE)
3. Root Mean Squared Error (RMSE)
4. R-Square (R^2 Score)

These metrics help determine whether the regression model is accurate and reliable.

Residual

- Actual – Predicted
- Also represented as y-yhat

$$\text{Residual} = \text{Actual Value} - \text{Predicted Value}$$

$$\text{Residual} = Y - \hat{Y}$$

Residual indicates how far the model's prediction is from the actual value.

1. A small residual means the prediction is close to the actual value.
2. A large residual means the prediction is inaccurate.

Therefore, a good regression model should have small residual values

Mean Squared Error (MSE)

Mean Squared Error (MSE) is the average of the squared residuals.

Residuals can be positive or negative.

If we simply add them, they may cancel each other. (-3 and +3) errors exist, the total becomes zero.

To avoid this problem, we square every residual before calculating the average.

Interpretation

Smaller MSE → Better regression model

Larger MSE → Poor regression model

$$MSE = \frac{1}{n} \sum (Y - \hat{Y})^2$$

Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is the square root of the Mean Squared Error.

Interpretation

Lower RMSE → Better prediction accuracy

Higher RMSE → Poor prediction accuracy

$$RMSE = \sqrt{MSE}$$

R-Square (R^2 Score)

R-Square (Coefficient of Determination) measures how well the regression model explains the variation in the dependent variable.

Unlike MSE and RMSE, which measure prediction error, R^2 measures how well the model fits the data.

A value closer to 1 indicates that the model explains most of the variation in the data.

$$R^2 = 1 - \frac{SS_{Residual}}{SS_{Total}}$$

$$R^2 = 1 - \frac{\sum(Y - \hat{Y})^2}{\sum(Y - \bar{Y})^2}$$

where

$SS_{Residual}$ = Residual Sum of Squares (Unexplained Variation)

SS_{Total} = Total Sum of Squares (Total Variation)

GIVEN DATA (Training / Test Data)

X (Hours)	Y (Actual Marks)
1	42
2	47
3	61
4	69
5	78

REGRESSION EQUATION (Best Fit Line)

$$\hat{Y} = 9.4X + 31.2$$

MEAN OF ACTUAL VALUES

$$\bar{Y} = \frac{42+47+61+69+78}{5} = \frac{297}{5} = 59.4$$

X (Hours)	Y (Actual Marks)	$\hat{Y}$ (Predicted Marks) $= 9.4X + 31.2$	Residual ($Y - \hat{Y}$) (e)	Squared Residual ($Y - \hat{Y}$) ²	$Y - \bar{Y}$ (Deviation from Mean)	($Y - \bar{Y}$) ²
1	42	40.6	1.4	1.96	-17.4	302.76
2	47	50.0	-3.0	9.00	-12.4	153.76
3	61	59.4	1.6	2.56	1.6	2.56
4	69	68.8	0.2	0.04	9.6	92.16
5	78	78.2	-0.2	0.04	18.6	345.96
TOTAL (Σ)	297	296.0	0.0	13.60	0.0	897.20

$$MSE = \frac{\sum (Y - \hat{Y})^2}{n} = \frac{13.60}{5} = 2.72$$

$$RMSE = \sqrt{MSE} = \sqrt{2.72} = 1.65$$

$$\begin{aligned}
 R^2 &= 1 - \frac{SS_{Residual}}{SS_{Total}} \\
 &= 1 - \frac{13.60}{897.20} = 1 - 0.01516 = \mathbf{0.985}
 \end{aligned}$$

Metric	Value	Interpretation
MSE	2.72	Lower is better
RMSE	1.65	Lower is better
R^2	0.985	Closer to 1 is better



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Linear Regression works well only when the relationship between variables is linear (straight line).

However, many real-world datasets have nonlinear (curved) relationships.

Examples:

Study Hours vs Marks

Fertilizer vs Crop Yield

Age vs Income

Sales vs Advertisement Cost

If the data forms a curve, a straight line cannot fit it accurately.

Use Polynomial Regression when:

1. The scatter plot shows a curved pattern.
2. The relationship between variables is nonlinear.
3. Polynomial regression fits a curve instead of a straight line by adding higher-degree terms

The degree of a polynomial is the highest exponent (power) of the independent variable X .

Degree 1 (Linear Regression)

Equation:

$$y = mX + c$$

power of $X = 1$

Graph: Straight Line

Degree 2 (Quadratic Polynomial)

Equation:

$$y = c + mX + aX^2$$

Graph: Curved (Parabola)

power of $X = 2$

Degree 3 (Cubic Polynomial)

Equation:

Highest power = 3

Graph : create more complex curves. (S shape)

$$y = c + mX + aX^2 + bX^3$$

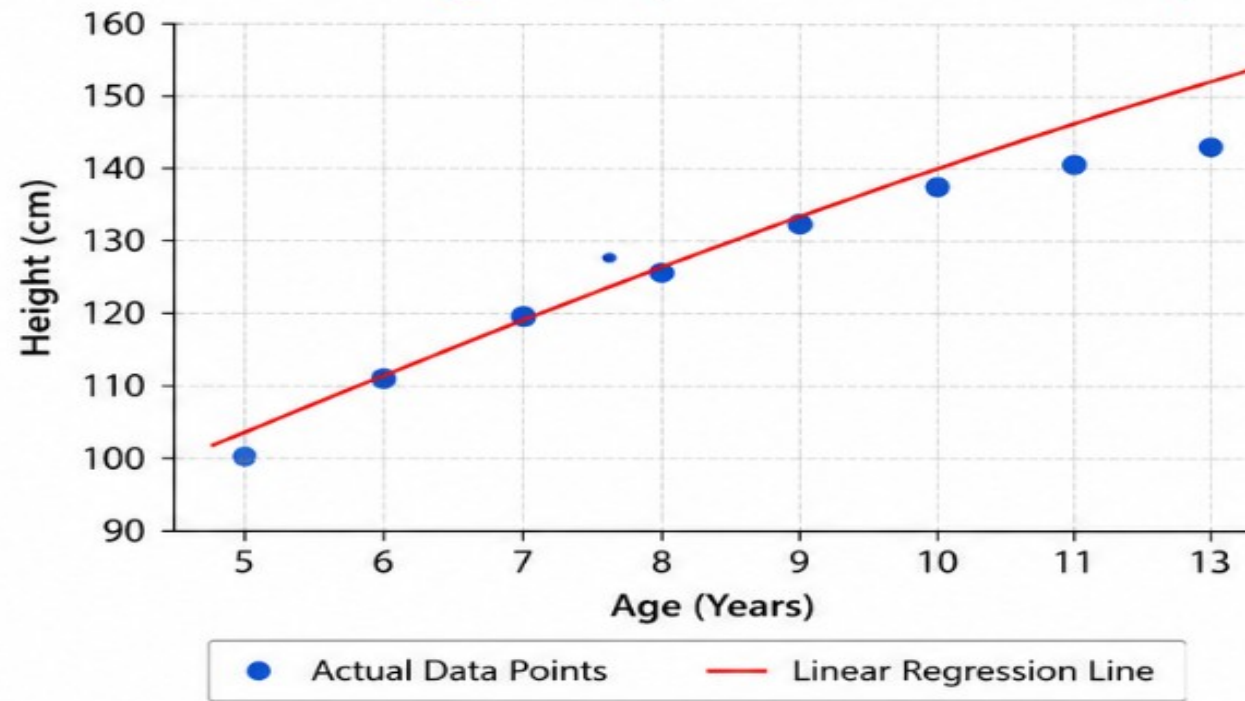
Linear Regression uses Least Squares to fit a straight-line equation.

Polynomial Regression also uses Least Squares, but it fits a polynomial equation.

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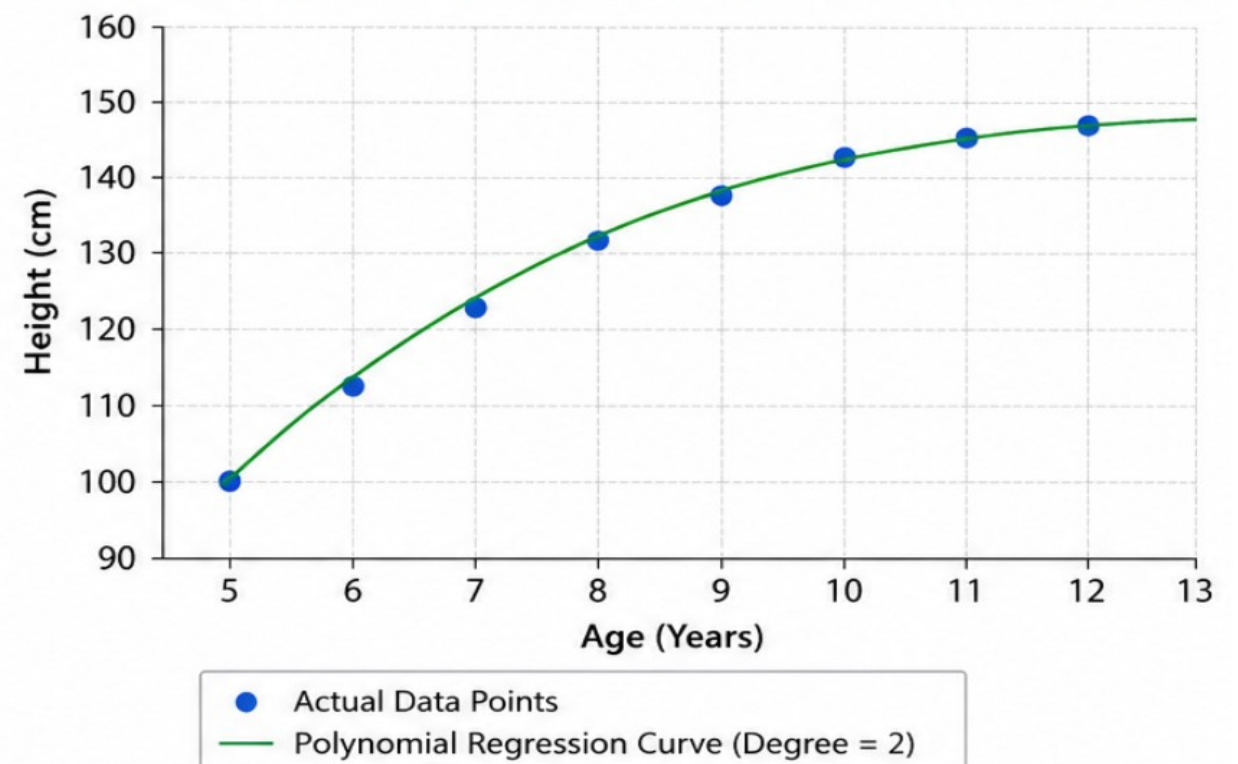
Polynomial Regression

1. Linear Regression (Poor Fit for Curved Data)



✗ The straight line cannot capture the curved pattern.

2. Polynomial Regression (Better Fit for Curved Data)



✓ The curve follows the actual pattern of the data.

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Polynomial Regression

Dataset

X	Y
1	6
2	11
3	18

Assume

$$y=c+mX+aX^2$$

First point

$$X=1$$

$$Y=6$$

$$6=c+m(1)+a(1)^2$$

$$6=c+m+a$$

Second point

$$X=2$$

$$Y=11$$

$$11=c+2m+4a$$

Third point

$$X=3$$

$$Y=18$$

$$18=c+3m+9a$$

$$6=c+m+a$$

$$11=c+2m+4a$$

$$18=c+3m+9a$$

These are three simultaneous equations.

By solving them,
we get
 $c=3$ $m=2$ $a=1$

So the polynomial equation becomes

$$y=3+2X+X^2$$

we use the Least Squares Method to solve three equations simultaneously.

Applications:

Financial market trends

Scientific experiments

Pros

- Fits curved relationships
- More flexible than linear

Cons

- Risk of overfitting
- Harder to interpret



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Problem Statement

A financial analyst wants to predict the closing price of a stock using its opening price, previous day's closing price, and trading volume.
You are provided with a historical stock dataset. Build a Linear Regression model to predict the closing price and evaluate its performance.

Dataset

Save the following as `stock_price_prediction.csv`

Open	Previous_Close	Volume	Close
100	98	150000	104
104	104	165000	106
106	106	170000	108
108	108	180000	110

Questions

1. Load the dataset using Pandas.
2. Display the first five records.
3. Identify the input features and target variable.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model.
6. Predict the closing prices.
7. Compare actual and predicted values.
8. Calculate MAE, MSE, RMSE and R^2 Score.
9. Plot Actual vs Predicted values.

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

Explanation

Pandas → Read and manipulate the dataset.

NumPy → Numerical operations.

Matplotlib → Plot graphs.

train_test_split → Split data into training and testing.

LinearRegression → Build the regression model.

metrics → Evaluate the model.

Stock_Price_Prediction

- stock_prediction.py
- stock_price_prediction.csv

$$Y = c + m_1X_1 + m_2X_2 + m_3X_3$$

Open (X_1)

Previous_Close (X_2)

Volume (X_3)

Close (Y)



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